AI-Assisted Proofs in Mathematical Risk Management: An Axiomatic Characterization of the Loss Ratio and a Proposed Reporting Standard

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Abstract

This paper has two aims of equal weight. The first is mathematical: to characterize the loss ratio as a risk indicator by necessary and sufficient conditions, as Cherny and Madan characterized performance measures such as the Sharpe ratio and the RAROC. The second is methodological: the result was obtained by putting AI to the test on this problem, with a panel of AI agents working under the author's direction, and the paper reports this process in full and proposes a standard for reporting AI-assisted proofs in mathematical risk management.

The loss ratio $\mathrm{LR}(X)$ of an insurance book X is the ratio of a loss measure $\rho(X)$, such as the expected loss or a tail measure, to a premium $\Pi(X)$. We show that both parts can be *implied* by the loss ratio itself. Along the cash line $c \mapsto \mathrm{LR}(X+c)$, obtained by adding sure losses c to the book, there is exactly one point where the ratio stops being finite; its location is the *implied premium*, the sure payment at which the book changes hands. The *implied risk measure* then follows from the accounting identity loss $=$ loss ratio $\times$ premium, and it is visible even where the ratio explodes, as the residue of the singularity. With this reading, axioms that refer only to values of the loss ratio – a point on each cash line at which the ratio is not finite, an identity for pooling comonotone books of zero premium (their underwriting results add), law invariance, and continuity (respectively monotonicity) of the sure amounts that remove the premium and the loss of a book – characterize the loss ratios $\mathrm{LR} = I_h/I_g$, where I_h, I_g are signed Choquet integrals with distortion functions h, g (respectively distortion premia), with (h, g) unique. The uniqueness of the singular point, the value one at sure losses, pooling at relative price, the unit-slope Möbius form of the cash lines, comonotonic additivity of the implied premium, and the association with a family of risk measures are derived rather than assumed. We show that no non-trivial loss ratio is monotone, show by counterexamples that none of the axioms can be dropped, and state candidly which parts of the result are classical.

The work used Claude (Anthropic) as a main assistant and to run panels of AI agents that surveyed the literature, proposed and ranked candidate theorems, drafted proofs and counterexamples, and reviewed the proofs adversarially; ChatGPT (OpenAI) and Gemini (Google) were used for an independent review of the proofs and for summaries of cited works. All decisions on content were the author's, and the central idea, the implied premium and risk measure, arose from the author's objections to AI-drafted formulations. We report the organization of the work, how the knowledge base was assembled, the errors in AI-generated mathematics that were found and corrected, the guardrails, a formal check in Lean of the steps specific to loss ratios, and what has not been verified, and distil these into a checklist for reporting AI-assisted proofs.

Keywords: loss ratio; implied premium; implied risk measure; axiomatic characterization; distortion premium; signed Choquet integral; comonotonic additivity; AI-assisted proofs; large language models; reporting standards; Lean.

*Department of Mathematics and Statistics, University College Dublin, and Model Library Ltd. *Draft of 8 November 2026.* The use of generative AI in this work is described in Section 12 and in the declaration before the references. The proofs have undergone AI-assisted critical review, which is fallible and does not constitute formal verification; the loss-ratio-specific steps have in addition been checked in Lean in an abstract setting. See Section 12.8.

1 Introduction

Like most mathematicians, the author has long carried a few questions that seemed worth answering. One of them, in actuarial science, is to characterize the loss ratio as a risk indicator by necessary and sufficient conditions, in the way that Cherny and Madan (2009) characterized performance measures such as the Sharpe ratio and the RAROC through their theory of acceptability indices. The loss ratio, claims divided by premium, is among the most widely used quantities in insurance underwriting, portfolio management and reinsurance pricing. Ex post it is incurred claims over earned premium; ex ante it is the ratio of a loss measure $\rho(X)$, such as an expected or a risk-adjusted loss, to a premium $\Pi(X)$ of the same position X. When the premium is a non-linear functional of the loss, or the numerator is a risk measure, it is not clear which functionals of a loss should be called a loss ratio, or which properties a loss ratio should have when books are pooled.

The question arose in risk reporting. Before the claims are known, a loss ratio is usually reported through its expected value, expected claims over premium. When the risk of a book is concentrated in the tail of its loss distribution, the expected loss ratio can be misleading: two books with the same expected loss ratio may differ greatly in their exposure to large losses. A natural remedy is to replace the mean of the claims by a tail measure, and to report, for instance, the value at risk or the tail value at risk (expected shortfall) of the claims divided by the premium (Artzner et al., 1999). The loss ratio then becomes a model-based, or mark-to-model, quantity: a risk measure of the loss, evaluated under a model of its distribution, divided by a premium principle. The ex-post accounting ratio is the special case in which the model is the realized outcome. Once such ratios are admitted, the question is which properties are necessary and sufficient for a functional of the loss to be a loss ratio of this kind. The tail value at risk and the value at risk are distortion functionals, so tail loss ratios over a distortion premium are covered by the results below (see the ratio $\mathrm{ES}_{0.9}/\mathbb{E}$ in Example 10.5).

This paper reports what happened when this question was put to AI. The author posed it to a panel of AI agents, in the manner of recent multi-agent efforts to obtain mathematical results with AI: agents surveyed ten strands of the literature, proposed candidate theorems, ranked them through judge agents, drafted proofs and counterexamples, and submitted the proofs to repeated adversarial review, while the author set the questions, rejected formulations and made every decision on content. The problem was chosen in part as a deliberate test. It is not a textbook exercise, its answer was not known in advance, and its solution needs machinery that is not routine, such as signed Choquet integrals. The paper therefore has two aims of equal weight: to answer the mathematical question, and to show, in a form that others can inspect, how AI was used to answer it.

The second aim makes transparency a requirement. A proof drafted by AI can be fluent and wrong, and the errors found in this project were mostly semantic: a correct argument for a slightly different statement, typically at a degenerate case, which careful reading can miss (Table 6). A reader therefore needs to know not only that AI was used, but for what, under whose direction, with which checks, which errors were found, and what has and has not been verified; the declarations of AI use that journals ask for record much less. Section 12 gives this account for the present paper and, in Section 12.9, distils it into a checklist for reporting AI-assisted proofs. To the author's knowledge, this is the first attempt to set such a standard in mathematical risk management and actuarial mathematics. It is offered as a proposal, for the community to test and improve.

We now describe the mathematical content. We work with the *extended loss ratio* $\mathrm{LR} = \rho/\Pi$, which allows negative values and zero premiums, defined abstractly by *association*: a family of risk measures $\{\rho_\beta\}$ is associated with LR if $\mathrm{LR}(X) = \beta \iff \rho_\beta(X) = 0$; the canonical family is $\rho_\beta = (\rho - \beta\Pi)/(1-\beta)$, $\beta \neq 1$. The present paper characterizes, by axioms stated on LR alone, the loss ratios that are ratios of two Choquet-type functionals.

Three features distinguish the problem from the classical characterizations of premium principles and risk measures (Yaari, 1987; Wang et al., 1997; Schmeidler, 1986; Wang et al., 2020a). First, the object is a ratio, while those results concern a single functional. Second, the ratio does not determine

its numerator and denominator: on positive losses, (ρ, Π) and $(\psi\rho, \psi\Pi)$ give the same ratio for every positive functional ψ. Third, the ratio is not monotone: we show that a loss ratio of two distortion functionals is monotone only if it is identically 1 (Proposition 4.23). The theory of acceptability indices of Cherny and Madan (2009), which rests on monotonicity, therefore does not apply, and the analogy that motivated the question has to be pursued by a different route.

Our approach is to recover the premium and the risk measure from the loss ratio. For a book X, consider the *cash line* $c \mapsto \mathrm{LR}(X+c)$. A premium is the sure amount at which a risk changes hands; on the cash line, a zero premium is a point at which the loss ratio is not finite. We define the *implied premium* $\Pi(X)$ as minus the unique such point, and the *implied risk measure* by the accounting identity $\rho(X) = \Pi(X)\mathrm{LR}(X)$, extended by cash shifts to books with zero premium. The implied risk measure is then minus the point at which the cash line vanishes (Corollary 4.14), and at books with zero premium it is the residue of the cash line at its singularity (Proposition 4.13). For a Bernoulli loss $\mathbf{1}_A$ with $P(A) = \frac{1}{4}$, the expected loss and the square-root distortion premium, the cash line is $(\frac{1}{4}+c)/(\frac{1}{2}+c)$; it is singular at $c = -\frac{1}{2}$, so $\Pi(\mathbf{1}_A) = \frac{1}{2}$ and $\rho(\mathbf{1}_A) = \frac{1}{2}\mathrm{LR}(\mathbf{1}_A) = \frac{1}{4}$ (Figure 1).

With this reading, Theorems 5.1 and 5.2 state the following. A map LR from bounded losses to $\mathbb{R} \cup \{\infty, \star\}$ is of the form I_h/I_g, with I_h, I_g signed Choquet integrals of distortions h, g of bounded variation, if and only if it satisfies the following axioms, stated in Section 4. Each refers only to values of LR. ($\mathsf{P}_\exists$) every cash line has a point where LR is not finite; that is, every book can be made unpriced by a sure payment. (R) if X, Y are comonotone and unpriced, $\mathrm{LR}(X) \notin \mathbb{R}$ and $\mathrm{LR}(Y) \notin \mathbb{R}$, then $u(\mathrm{LR}(X+Y+u)-1) = s(\mathrm{LR}(X+s)-1) + t(\mathrm{LR}(Y+t)-1)$ for all non-zero s, t, u, all three values being finite: written at the premium s, an unpriced book has the underwriting result $s(\mathrm{LR}(X+s)-1)$, and the underwriting results of comonotone books add. (LI) LR is law invariant. (RC) the sure amounts that make X unpriced and that remove its loss depend uniformly continuously on X. If (RC) is replaced by (RM), monotonicity of these two amounts in X, then h and g are distortion functions and no continuity assumption is needed. In both cases (h, g) is unique, $\Pi = I_g$ and $\rho = I_h$. No axiom refers to a premium, a risk measure or a weight. The following are consequences: (P) each cash line has exactly one non-finite point; (N) $\mathrm{LR}(c) = 1$ for sure losses $c \neq 0$; (CM) the loss ratio of a pool of comonotone books is the combination of their loss ratios weighted by their relative prices; and, further, the unit-slope Möbius form of the cash lines, comonotonic additivity of the implied premium, scale invariance and the association.

Axioms ($\mathsf{P}_\exists$) and (R) are equivalent to (P), (N) and (CM) together, and, given (P) and (N), (CM) is equivalent to comonotonic additivity of the implied premium and risk measure, the axiom of Yaari (1987) and Wang et al. (1997); once this equivalence is established, the representation step follows from the theory of comonotonic additive functionals (Schmeidler, 1986; Murofushi et al., 1994; Cerreia-Vioglio et al., 2012; Wang et al., 2020a). The contributions specific to loss ratios are: the construction of the implied premium and risk measure; the proof that the pooling identity yields additivity of the implied pair on all comonotone pairs, including books that break even, on which a premium given exogenously would not be identified (Lemma 7.5 and Remark 4.11); unconditional uniqueness of (h, g); the impossibility of a monotone loss ratio and its consequences for the associated family (Corollary 10.1); a version for nonnegative losses (Theorem 8.3); and counterexamples showing that none of the axioms ($\mathsf{P}_\exists$), (R), (LI), (RC), (RM) can be dropped, together with a proof that no pooling axiom using only the finite values of LR can replace (R).

On the side of AI, the outcome of the experiment was as follows. The work was done with Claude (Anthropic), used as a main assistant and to coordinate panels of AI agents on designated parts of the project: a roundtable of six agents and a larger panel covering ten literature strands surveyed the literature, proposed candidate statements, ranked them by judge agents, drafted proofs and submitted them to repeated AI-assisted critical review; the main assistant constructed examples and counterexamples and drafted and revised the manuscript. ChatGPT (OpenAI) was used for an independent informal review of the proofs, and Gemini (Google) for summaries of key cited works. The agents produced a complete if-and-only-if theorem with proofs, but in a formulation that assumed a cash-line axiom

and so supplied ρ and Π by fiat. The formulation through the implied premium and the implied risk measure, which is the central idea of the paper, arose from the author's objections to these AI-drafted versions. The AI-generated material was treated as proposals throughout. The reviews found seventeen errors in AI-generated mathematics, including a wrong sufficient condition for quasi-convexity and an incorrect uniqueness claim, all of which were corrected (Table 6); three of them had been missed by the earlier checks and were found by the review with a model from a different provider. The numbers in the examples were recomputed; the statements taken from three of the main sources were checked against those papers; and the steps specific to loss ratios were formalized in the proof assistant Lean and checked by its kernel, with the classical representation step taken as a hypothesis. In an earlier version the pooling and regularity axioms were stated through the implied premium and risk measure; following a reviewer's remark, the author required that every axiom refer only to values of the loss ratio, and a panel of nine agents, with an adversarial check of each proposal, produced the axioms ($\mathsf{P}_\exists$) and (R) used here (Section 12.7).

The paper is organized as follows. Section 2 reviews the related literature, and Section 3 introduces the extended loss ratio. Section 4 defines the implied premium and risk measure and states the axioms. Section 5 contains the main theorems and examples, and Section 6 discusses the role of the axioms. Proofs are in Section 7. Section 8 treats nonnegative losses, Section 9 gives the counterexamples, and Section 10 derives consequences. Section 11 lists open problems. Section 12 reports the use of AI: the organization of the work, the knowledge base, the errors found, the guardrails, the formal check in Lean (Section 12.6), the verification status, and the proposed reporting standard (Section 12.9). Readers mainly interested in the use of AI can read Section 12 directly after this introduction; it refers to the mathematical sections only for the errors found and for the results checked in Lean.

2 Related literature and what is different here

The loss ratio touches four literatures: the axiomatic theory of premia and risk measures, the theory of performance ratios and acceptability indices, implicitly defined risk functionals, and ratio representations in decision theory and statistics. For each we say what is known, what we use, and where our problem differs. Section 2.5 collects the differences in a table.

2.1 Distortion premia, Choquet integrals and signed Choquet integrals

The axiomatic theory of insurance premia starts from properties of a single premium functional H on losses: translation invariance, $H(X+c)=H(X)+c$, and "no unjustified loading", $H(c)=c$ for constant losses (Gerber, 1974), monotonicity, law invariance, and, most importantly, additivity for comonotonic risks. The dual theory of Yaari (1987) replaces utility transformations of payoffs by distortions of probabilities, which leads to the Choquet-type representation $\int g(P(X>x))\,dx$ and to additivity for comonotonic risks, and Wang (1996) introduced it as a premium principle. Wang et al. (1997) characterize the *distortion premium* C_g by law invariance, monotonicity, comonotonic additivity, a continuity condition and $H(1)=1$. The mathematical backbone is Schmeidler's (1986) theorem: a monotone, comonotonic additive functional on bounded measurable functions is a Choquet integral with respect to a capacity, with no continuity needed; see also Greco (1982) and Denneberg (1994) on cones. Coherent risk measures are defined by monotonicity, translation invariance, subadditivity and positive homogeneity (Artzner et al., 1999; Delbaen, 2002); in the law-invariant case, comonotonic coherent risk measures are mixtures of expected shortfalls over levels in $[0,1]$ (Kusuoka, 2001; Jouini et al., 2006; Föllmer and Schied, 2016).

Without monotonicity the corresponding representation theory is that of *signed* Choquet integrals. Murofushi et al. (1994) and Cerreia-Vioglio et al. (2012, 2015) show that comonotonic additive functionals of bounded variation are exactly differences of Choquet integrals with respect to set functions of bounded variation. Wang et al. (2020a) prove the law-invariant version used here: on an atomless

L^∞, a functional is law invariant, comonotonic additive and uniformly norm continuous if and only if it is a signed Choquet integral I_h with h of bounded variation. They also show that I_h is convex if and only if h is concave, and Wang and Wei (2020) study when level sets of such functionals are convex. Wang et al. (2020b) extend the characterization to law-invariant convex cones containing L^∞, such as L^p, by adding continuity at infinity, which holds automatically on L^∞, and give a representation of convex signed Choquet integrals by mixtures of expected shortfalls.

These theorems underlie our representation step, and Theorems 5.1 and 5.2 are the loss-ratio counterparts of Wang et al. (2020a, Thm. 1) and of Wang et al. (1997). We re-prove the representation directly (Steps 5–8 of Section 7) to keep track of constants and because we need no continuity in the monotone tier.

All of these results characterize one functional by axioms placed on that functional. A loss ratio is a ratio of two, and the axioms must be placed on the ratio, which does not determine its parts (Section 1). Bridging this gap is the content of Sections 4 and 7. We are not aware of an if-and-only-if characterization of a ratio of two Choquet integrals, or of two risk measures, in this literature.

2.2 Performance ratios and acceptability indices

The best-known ratios of risk functionals are performance measures. The gain–loss ratio of Bernardo and Ledoit (2000) divides the expected positive part of a payoff by the expected negative part, and Biagini and Pinar (2013) study its optimization. Cherny and Madan (2009) give an axiomatic theory of *acceptability indices*: a map α that is monotone, quasi-concave, scale invariant and Fatou is characterized by an increasing family of coherent risk measures $\{\rho_x\}$ with $\alpha(X) = \sup\{x : \rho_x(X) \le 0\}$, and the family is recovered by the cash-shift formula $\rho_x(X) = \inf\{c : \alpha(X + c) \ge x\}$. Dynamic versions and the correspondence between indices and families are developed by Bielecki et al. (2014) and Biagini and Bion-Nadal (2014), and the correspondence between quasi-convex functionals and acceptance families by Drapeau and Kupper (2013). The Sharpe ratio is not an index in this sense, and the conic-finance programme of Cherny and Madan (2010) uses the *spread* between ask and bid prices instead of a ratio. Cheridito and Kromer (2013) propose reward–risk ratios, a reward functional divided by a risk functional, and list axioms that make such a ratio a performance measure, giving sufficient conditions rather than a characterization.

Han et al. (2025) give an axiomatic theory of diversification indices. They note that the diversification ratio $\phi(\sum_i X_i)/\sum_i \phi(X_i)$ of a risk measure ϕ may be negative, needs the conventions $0/0 = 0$ and $1/0 = \infty$, and is not location invariant, and they characterize instead the *diversification quotient* α^*/α, where α^* is the level at which a decreasing family (ρ_β) of risk measures, evaluated at the pooled portfolio, reaches $\sum_i \rho_\alpha(X_i)$. Six axioms (non-negativity, location and scale invariance, rationality, normalization and continuity) characterize the quotients built from monotone, cash-additive and positively homogeneous families (Han et al., 2025, Thm. 1), and portfolio convexity restricts the class to coherent families when there are at least four assets (Han et al., 2025, Thm. 2); they also relate the quotient to the acceptability indices of Cherny and Madan (2009). This is the closest if-and-only-if result of association type. It shares two features with ours: a ratio-type quantity is recovered as the level at which a family of risk measures crosses a threshold (compare Corollary 10.1), and the naive ratio needs conventions where its denominator vanishes (compare the labels ∞ and $\star$). It differs in the object, an index on vectors of risks that is invariant under location and scale, whereas a loss ratio is a function of a single book that moves along each cash line as a Möbius map; and in monotonicity, which their rationality axiom uses and which a loss ratio cannot have (Proposition 4.23).

The cash-shift recovery formula of Cherny and Madan (2009) and its generalizations is the model for our reading of ρ and Π from cash lines, and the association is the loss-ratio analogue of the level-crossing definition of an index. Three differences matter.

First, *direction*. In an acceptability index, larger gains are better, so the index is monotone and the family $\{\rho_x\}$ is monotone in the level. In a reward–risk ratio, the numerator is a reward and the

denominator a risk, so the ratio is monotone again. In a loss ratio, numerator and denominator are both increasing in the loss. We show that the ratio is then not monotone (Proposition 4.23), and that the associated family $\rho_\beta = (\rho - \beta\Pi)/(1-\beta)$ is not monotone either (Corollary 10.1): it is a *signed* Choquet integral even when ρ and Π are distortion premia. The index theorems therefore do not apply, and the signed tier of Theorem 5.1 is not a luxury but the natural home of the associated family. Second, *cardinality*. Indices are ordinal objects, determined up to increasing transformations, whereas the loss ratio is cardinal: it obeys an accounting identity under pooling (Section 4.4). Third, *what is characterized*. We characterize the ratio together with its numerator and denominator, not an index through a family.

2.3 Level-crossing and implicitly defined functionals

Several risk functionals are defined implicitly, by where a monotone family crosses a level. Orlicz premia solve $\mathbb{E}[\varphi(X/\pi)] = 1-\alpha$ for π (Haezendonck and Goovaerts, 1982; Bellini and Rosazza Gianin, 2008), and PELVE is the level at which expected shortfall equals a given value-at-risk (Li and Wang, 2023; Fiori and Rosazza Gianin, 2023). Return risk measures (Bellini et al., 2018) and star-shaped risk measures (Castagnoli et al., 2022) replace cash additivity by positive homogeneity or star-shapedness. The association belongs to this family: it defines $\mathrm{LR}(X)$ as the level β at which a cash-additive family ρ_β vanishes.

The results in this literature establish *existence, uniqueness and comparative statics* of the implicit functional. They do not represent it as a ratio of two explicit functionals. In our setting the implicit definition is recovered as a corollary (Corollary 10.1), and it holds exactly off the set $\{\rho = \Pi = 0\}$, not on all of L^∞, a point that the implicit definition alone obscures.

2.4 Ratio representations in decision theory and statistics

In decision theory, Chew (1983) shows that weighted-utility preferences are represented by a *ratio* of two linear functionals, while Dekel (1986) and Fishburn (1983) show that the weaker *betweenness* property yields only an implicit, ordinal representation. This distinction is exactly the one we meet: a mediant inequality for pooled loss ratios is a betweenness property and cannot characterize the ratio, whereas the premium-weighted identity is the weighted-utility analogue and can (Section 4.4 and the cube witness in Section 9). In statistics, Gneiting (2011) shows that ratios of expectations $\mathbb{E}[r(Y)]/\mathbb{E}[s(Y)]$ are elicitable, which is the only published if-and-only-if statement about a ratio functional that we know of; it concerns forecasting, not pricing, and has no counterpart of pooling or cash shifts.

2.5 Summary of the differences

None of our proofs depends on a citation: every load-bearing step is proved in the text.

3 The extended loss ratio

3.1 Values and cash lines

Throughout, $(\Omega, \mathscr{F}, P)$ is an atomless probability space and L^∞ is the space of essentially bounded random variables modulo null sets, with the norm $\|\cdot\|_\infty$. Elements of L^∞ represent losses, so positive values are losses and negative values are gains.

Let $\widehat{\mathbb{R}}_\star := \mathbb{R} \cup \{\infty\} \cup \{\star\}$. For reals r, s put

$$[r:s] := \begin{cases} r/s & s \neq 0, \\ \infty & s = 0 \neq r, \\ \star & r = s = 0. \end{cases}$$

Literature	What it characterizes	Difference from this paper
Yaari (1987); Wang, Young, Panjer (1997)	One premium functional H; axioms on H (law invariance, monotonicity, comonotonic additivity).	We characterize a *ratio* of two such functionals with axioms on the ratio. Monotonicity is placed on the *implied premium and risk measure*, never on the ratio.
Schmeidler (1986); Murofushi–Sugeno–Machida (1994); Cerreia-Vioglio et al. (2012, 2015); Wang–Wei–Willmot (2020)	Representation of one comonotonic additive functional as a (signed) Choquet integral.	Used as the engine (applied to ρ and Π). New: the axioms are on LR, comonotonic additivity of ρ and Π is *derived* (also on the break-even set), and uniqueness is unconditional.
Cherny–Madan (2009); Bielecki–Cialenco–Zhang (2014); Biagini–Bion-Nadal (2014)	Monotone, quasi-concave indices $\Leftrightarrow$ increasing families of coherent measures.	LR cannot be monotone (Prop. 4.23); the associated family is signed and non-monotone. Cardinal (pooling identity) rather than ordinal.
Cherdito–Kromer (2013)	Axioms for reward–risk ratios (sufficient conditions).	Reward over risk is monotone; loss over premium is not. We give an *iff*, within a stated class.
Han–Lin–Wang (2025)	Diversification quotient: iff axiomatization of a level-crossing index.	Closest in spirit; monotone families and a portfolio index. We characterize the ratio and its parts.
Haezendonck–Goovaerts (1982); Li–Wang (2023); Bellini et al. (2018)	Implicitly defined premia/risk measures, return risk measures.	Existence and uniqueness of an implicit functional, not a ratio representation. The association is recovered as a corollary and shown to hold only off $\{\rho = \Pi = 0\}$.
Chew (1983); Dekel (1986)	Ratio (weighted utility) versus implicit (betweenness) representations.	Explains why a mediant inequality is not enough and a premium-weighted identity is needed.
Gneiting (2011)	Elicitability of ratios of expectations.	Statistical, no axioms on pooling or cash shifts.

Table 1: Positioning of the paper. "Iff" means a characterization by necessary and sufficient axioms.

The symbol ∞ records "non-zero loss, zero premium" and $\star$ records the indeterminate $0/0$. Only the distinction between finite and non-finite values matters for the theory; the two labels are a convenience (Remark 4.18).

Definition 3.1 (Extended loss ratio, cash line, finite domain)**.** An *extended loss ratio* is a map $\mathrm{LR} : L^\infty \to \widehat{\mathbb{R}}_\star$. Its *cash line* at X is $\phi_X(c) := \mathrm{LR}(X+c)$ for $c \in \mathbb{R}$. Its *finite domain* is $D := \{X : \mathrm{LR}(X) \in \mathbb{R}\}$.

Evaluating LR at cash-shifted positions $X+c$, including negative c, is required by the associated family, since ρ_β is defined through $\mathrm{LR}(X-m) = \beta$. Allowing $c < 0$ is therefore part of the framework rather than an additional assumption. Section 8 treats the actuarial case of nonnegative losses where only $c \geq 0$ is available.

For instance, if $X = \mathbf{1}_A$ with $P(A) = p$, the ratio of the expected loss to the premium $\int_0^\infty \sqrt{P(X > x)}\,dx$ has cash line $c \mapsto (p+c)/(\sqrt{p}+c)$; Example 4.17 and Figure 1 discuss this case.

3.2 Choquet integrals and distortions

For a bounded Borel function $k:[0,1]\to\mathbb{R}$ with $k(0)=0$ define the *(signed) Choquet integral* of a loss $X\in L^\infty$ by

$$I_k(X) := \int_0^\infty k\big(P(X>x)\big)\,dx + \int_{-\infty}^0 \Big[k\big(P(X>x)\big)-k(1)\Big]\,dx.$$

Write $\mathrm{TV}(k)$ for the total variation of k and let

$$\mathscr{H} := \{k:\ \mathrm{TV}(k)<\infty,\ k(0)=0\}, \qquad \mathscr{H}_1 := \{k\in\mathscr{H}:\ k(1)=1\}.$$

A *distortion function* is a non-decreasing element of $\mathscr{H}_1$; jumps are allowed, so value-at-risk-type distortions are included. On nonnegative losses we write $C_k := I_k$. Three facts are used throughout.

- $I_k(X+c)=I_k(X)+c\,k(1)$ (cash slope $k(1)$), and I_k is law invariant.
- For $k\in\mathscr{H}$, I_k is the signed Choquet integral of Wang et al. (2020a, Def. 1), and it is Lipschitz with constant $\mathrm{TV}(k)$ (Wang et al., 2020a).
- I_k is monotone if and only if k is non-decreasing (Wang et al., 2020a), and I_k is convex if and only if k is concave (Wang et al., 2020a, Thm. 3).

Two random variables X,Y are *comonotone* if $(X(\omega)-X(\omega'))(Y(\omega)-Y(\omega'))\geq 0$ for $P\otimes P$-almost every (ω,ω'). Constants are comonotone with everything. A functional F is *comonotonic additive* if $F(X+Y)=F(X)+F(Y)$ whenever X,Y are comonotone. Every I_k with $k\in\mathscr{H}$ is comonotonic additive (proof in Section 7).

4 The axioms, the implied premium and the implied risk measure

In this section we state the axioms and define the premium and the risk measure implied by a loss ratio. Every axiom refers only to values of LR: it evaluates LR at books and at their cash shifts, and uses only arithmetic and whether a value is finite. No axiom refers to a premium, a risk measure, a weight, or a point of a cash line defined from LR. There are five axioms: ($\mathsf{P}_\exists$) and (R) in Section 4.1, (LI) in Section 4.6, and the regularity axioms (RC) and (RM) in Section 4.7. Axioms ($\mathsf{P}_\exists$) and (R) imply three properties, (P), (N) and (CM), which are equivalent to them and which give the axioms their meaning: a unique price point on each cash line, a loss ratio of one for sure losses, and comonotone pooling at relative price (Proposition 4.5). The implied premium and the implied risk measure are minus the price point and minus the zero of the cash line (Definition 4.8 and Corollary 4.14). They are used to interpret the axioms and in the proofs. Table 2 at the end of the section summarizes the axioms and the derived properties.

4.1 Axioms ($\mathsf{P}_\exists$) and (R): pooling of books with zero premium

A premium is the sure amount at which a risk changes hands. A book whose loss ratio is not finite is one that changes hands at zero premium: it has been transferred together with exactly its premium and commands no further premium. We call such a book *unpriced*.

Axiom 4.1 (($\mathsf{P}_\exists$): every book can be made unpriced)**.** For every $X\in L^\infty$ there is $c\in\mathbb{R}$ with $\mathrm{LR}(X+c)\notin\mathbb{R}$.

Axiom 4.2 ((R): comonotone pooling of unpriced books)**.** Let $X,Y\in L^\infty$ be comonotone with $\mathrm{LR}(X)\notin\mathbb{R}$ and $\mathrm{LR}(Y)\notin\mathbb{R}$. Then for all $s,t,u\in\mathbb{R}\setminus\{0\}$ the values $\mathrm{LR}(X+s)$, $\mathrm{LR}(Y+t)$ and $\mathrm{LR}(X+Y+u)$ are real, and

$$u\big(\mathrm{LR}(X+Y+u)-1\big) = s\big(\mathrm{LR}(X+s)-1\big)+t\big(\mathrm{LR}(Y+t)-1\big). \tag{1}$$

Axiom (R) is the accounting identity "premium $\times$ (loss ratio -1) = underwriting result", stated in loss ratios alone. If X is unpriced, adding a sure loss $s \neq 0$ creates a book whose premium is s, since sure amounts are priced at face value, and $s(\mathrm{LR}(X+s)-1)$ is the underwriting result of that book, claims minus premium, in units of cash. Axiom (R) says two things. First, the underwriting result does not depend on the sure amount added, since a sure loss carries no underwriting result: this is (1) with $Y=0$, once it is known that 0 is unpriced and $\mathrm{LR}(t)=1$, both of which (R) implies (Proposition 4.5). Second, the underwriting results of comonotone books, which cannot hedge each other, add up when the books are pooled, and a pool of unpriced books is again unpriced. The hypothesis that X and Y are unpriced is a statement about values of LR; it only selects the books to which the identity applies. A general book enters through its cash shifts. Taking $u=s+t\neq 0$ in (1) gives

$$\mathrm{LR}(X+Y+s+t)=\frac{s}{s+t}\,\mathrm{LR}(X+s)+\frac{t}{s+t}\,\mathrm{LR}(Y+t)\,: \tag{2}$$

two comonotone books written at the premiums s and t pool to the average of their loss ratios weighted by these premiums. This is pooling at relative price, with observed cash premiums as weights.

The hypothesis that X and Y are unpriced cannot be removed, and it cannot be replaced by a condition on finite values of LR. For a loss ratio LR write $\widetilde{\mathrm{LR}}$ for the real-valued map equal to LR where LR is finite and to 1 elsewhere. A condition *depends on* LR *only through its finite values* if it holds for LR whenever it holds for some LR° with $\widetilde{\mathrm{LR}^\circ}=\widetilde{\mathrm{LR}}$. Conditions on limits of finite values along cash lines, such as $\lim_{s\to 0} s(\mathrm{LR}(X+s)-1)$ or $\lim_{|s|\to\infty} s(\mathrm{LR}(X+s)-1)$, are of this kind, since under (P) the values $\mathrm{LR}(X+s)$ near the singular point are finite.

Proposition 4.3 (Non-finite values are needed)**.** *Let* $\Pi'=\max(I_{\sqrt{t}}, I_{\min(1,1.5t)})$ *and* $\mathrm{LR}'=[\Pi':\Pi']$. *Then* LR' *satisfies* ($\mathsf{P}_\exists$), (P), (N), (LI), (RC) *and* (RM)*, its finite values are all equal to* 1*, and* LR' *is not of the form* $[I_h:I_g]$ *with* $h,g\in\mathscr{H}_1$*. Consequently, no condition that depends on* LR *only through its finite values and holds for every loss ratio* $[I_h:I_g]$ *can replace* (R) *in Theorems 5.1 and 5.2.*

Proof. Π' is law invariant, monotone, 1-Lipschitz and cash additive with $\Pi'(c)=c$, so the cash line of X under LR' equals 1 except at $c=-\Pi'(X)$, where it is not finite. This gives ($\mathsf{P}_\exists$), (P), (N) and (LI). The sure amount a with $\mathrm{LR}'(X-a)\notin\mathbb{R}$ is $\Pi'(X)$; a book W never has $\mathrm{LR}'(W)=0$, and $\mathrm{LR}'(W)\notin\mathbb{R}$ with $\mathrm{LR}'(W+1)=1$ holds exactly when $\Pi'(W)=0$. Both sure amounts in (RC) and (RM) are therefore $\Pi'(X)$, and both axioms hold. If $\mathrm{LR}'=[I_h:I_g]$, the two maps have the same non-finite books, so $I_g(X)=\Pi'(X)$ for every X by (P), and Π' would be comonotonic additive; Example 9.4 shows that it is not. Finally, $\widetilde{\mathrm{LR}'}\equiv 1=\widetilde{\mathrm{LR}^\circ}$ for $\mathrm{LR}^\circ=[\mathbb{E}:\mathbb{E}]$, which is of the form $[I_h:I_g]$; so every condition of the stated kind holds for LR'. □

On books that break even, the finite values of the cash line carry no information about the premium, and only the location of the non-finite values does (Proposition 4.15). The following remark shows the same for the limit at a vanishing premium.

Remark 4.4 (The limit at zero premium). For $\mathrm{LR}=[I_h:I_g]$, $s(\mathrm{LR}(X+s)-1)=s(I_h(X)-I_g(X))/(I_g(X)+s)$. Its limit as $s\to 0$ exists for every X: it is $I_h(X)$ if X is unpriced and 0 otherwise. An axiom requiring that this limit exist is therefore automatic, and the limit is 0 on unpriced books with $I_h(X)=0$ as well as on all books with a premium, so it does not identify the unpriced books. Nor can (1) be stated for the limits on all comonotone pairs: take $\rho=\mathbb{E}$, $\Pi=I_{\sqrt{t}}$, $X=\mathbf{1}_A$ with $P(A)=\frac{1}{4}$ and $Y=-\frac{1}{2}$. The limits for X and Y are 0, since $\Pi(X)=\frac{1}{2}$ and Y is a sure amount, while $X+Y$ is unpriced and its limit is $\mathbb{E}(X+Y)=-\frac{1}{4}$. The hypothesis of (R) accounts for such pools, whose premiums cancel.

Proposition 4.5 (The derived properties)**.** *Axioms* ($\mathsf{P}_\exists$) *and* (R) *hold if and only if the properties* (P)*,* (N) *and* (CM) *below hold.*

The proof is in Section 7. The rest of this section states the three properties, reads off the implied premium and risk measure, and explains their meaning.

4.2 The price point, the zero and the implied premium

Property 4.6 ((P): one price point on each cash line)**.** For every $X \in L^\infty$ there is exactly one $c \in \mathbb{R}$ at which $\mathrm{LR}(X+c) \notin \mathbb{R}$.

Definition 4.7 (Price point and zero of a cash line)**.** Under (P), the *price point* of X is the unique $c_X^\infty \in \mathbb{R}$ at which $\mathrm{LR}(X+c_X^\infty)$ is not finite. If the cash line of X has at most one zero, its *zero* c_X^0 is the unique $c \in \mathbb{R}$ with $\mathrm{LR}(X+c)=0$ if there is one, and $c_X^0 := c_X^\infty$ if the cash line has no zero.

Both points are read off the cash line: c_X^∞ is where the loss ratio explodes and c_X^0 is where it vanishes. They are the cash offsets at the levels $\beta=\infty$ and $\beta=0$, that is, the sure amounts c with $\mathrm{LR}(X+c)=\beta$; this is the analogue for loss ratios of the recovery $\rho_x(X)=\inf\{c : \alpha(X+c)\geq x\}$ of a family of risk measures from an acceptability index α (Cherny and Madan, 2009). The convention for a cash line without a zero is explained after Proposition 4.13: under the axioms, such a cash line equals 1 away from its price point. Since the cash line of $X+c$ is the cash line of X shifted by c,

$$c_{X+c}^\infty = c_X^\infty - c, \qquad c_{X+c}^0 = c_X^0 - c \qquad \text{for all } X \in L^\infty,\ c \in \mathbb{R}. \tag{3}$$

Definition 4.8 (Implied premium)**.** Under (P), the *implied premium* of X is $\Pi(X) := -c_X^\infty$, minus the price point of X.

By (3),

$$\Pi(X+c) = \Pi(X) + c \qquad \text{for all } X \in L^\infty,\ c \in \mathbb{R}, \tag{4}$$

and X belongs to the finite domain $D=\{\mathrm{LR} \in \mathbb{R}\}$ if and only if $c_X^\infty \neq 0$, that is, $\Pi(X) \neq 0$. Property (P) does not distinguish between the two non-finite values; they play the role of labels (Remark 4.18).

The definition formalizes the notion of a price. A premium is the sure amount at which a risk changes hands: once the insurer holds X and has received $\Pi(X)$, the position $X-\Pi(X)$ commands no further premium, and on the cash line a zero premium is exactly a point where the loss ratio is not finite. In terms of the canonical family $\rho_\beta = (\rho-\beta\Pi)/(1-\beta)$, for books X outside the break-even set, $\rho_\beta(X)$ is the sure offset m at which $\mathrm{LR}(X-m)=\beta$ (Corollary 10.1(b)), and $\rho_\beta \to \Pi$ as $\beta \to \infty$; the implied premium is therefore the member of the associated family at level ∞, in the same way that Cherny and Madan (2009) recover a family of risk measures from an acceptability index α through $\rho_x(X) = \inf\{c : \alpha(X+c)\geq x\}$. The definition also agrees with ratemaking practice, in which the premium is obtained from a loss cost and a permissible loss ratio (Werner and Modlin, 2016). Finally, the price point measures the premium in units of cash, which removes the indeterminacy $(\rho,\Pi) \mapsto (\lambda\rho,\lambda\Pi)$ of the ratio. A shift $X+c$ with $c<0$ is the book transferred together with a sure payment $|c|$.

Property (P) is ordinal: it is unchanged if LR is replaced by $\psi\circ\mathrm{LR}$ for any $\psi:\mathbb{R}\to\mathbb{R}$, the non-finite values being kept. It excludes ratios that are finite along a whole cash line, such as $\arctan(\rho/\Pi)$, and ratios with several singular points.

4.3 Property (N): a sure loss is priced at cost

Property 4.9 ((N))**.** $\mathrm{LR}(c)=1$ for every constant $c \neq 0$.

A sure loss carries no risk, so its fair premium is its face value; this is the "no unjustified loading" property of premium principles (Gerber, 1974). Under (P), property (N) implies $\Pi(c)=c$ for all c: the cash line of 0 equals 1 away from 0, so $\Pi(0)=0$, and $\Pi(c)=c$ by (4). Premiums with a proportional loading, such as $(1+\theta)\mathbb{E}$, violate (N).

4.4 Property (CM): comonotone pooling at relative cost

For $X,Y \in D$ with $c_X^\infty + c_Y^\infty \neq 0$ call

$$w_{X,Y} := \frac{c_X^\infty}{c_X^\infty + c_Y^\infty}, \qquad w_{Y,X} = 1 - w_{X,Y} = \frac{c_Y^\infty}{c_X^\infty + c_Y^\infty},$$

the *relative cost* of X and of Y, respectively. Since $-c_X^\infty$ is the sure payment at which X changes hands, $w_{X,Y}$ is the relative price of X in the comonotone portfolio $\{X,Y\}$: the price of X as a fraction of the prices of the two books together, read off their cash lines. In terms of the implied premium, $w_{X,Y} = \Pi(X)/(\Pi(X)+\Pi(Y))$.

Property 4.10 ((CM))**.** For comonotone $X,Y \in D$ with $c_X^\infty + c_Y^\infty \neq 0$,

$$\mathrm{LR}(X+Y) = w_{X,Y}\,\mathrm{LR}(X) + w_{Y,X}\,\mathrm{LR}(Y).$$

Comonotone books do not hedge each other, so the cost of the pooled book should be the sum of the costs of its parts, and then $w_{X,Y}$ is the share of X in the cost of the pool. This is a consequence of (CM) and not part of it: it is the comonotonic additivity $\Pi(X+Y) = \Pi(X)+\Pi(Y)$ of Lemma 7.5. Property (CM) then says that the loss ratio of a comonotone pool is allocated to its constituents according to relative cost: the pooled loss ratio is the combination of the two loss ratios whose weights are their relative costs. Since $\rho = \Pi \cdot \mathrm{LR}$, this is the accounting statement that the loss of the pool is the sum of the losses of the parts and the premium of the pool is the sum of their premiums, so that the loss ratio of the pool is total loss over total premium. For books that are not comonotone, diversification will in general make the pooled premium smaller than the sum of the premiums; this is why Yaari (1987) and Wang et al. (1997) impose additivity only for comonotone risks.

The weights always add up to one. They lie in $[0,1]$, and the combination is convex, exactly when $\Pi(X)$ and $\Pi(Y)$ have the same sign, in particular when both premiums are positive, which is the case of practical interest. Premiums of opposite sign do occur, since $\Pi(c) = c$ for sure losses and gains, and then one weight is negative and the other exceeds one, so that the combination is affine and not convex.

A constant is comonotone with every book. Taking $Y = c$ in (CM) and using (N) and $\Pi(c) = c$ gives, for $X \in D$, $c \neq 0$ and $c \neq -\Pi(X)$,

$$\mathrm{LR}(X+c) = \frac{\Pi(X)}{\Pi(X)+c}\,\mathrm{LR}(X) + \frac{c}{\Pi(X)+c}\cdot 1 = \frac{\Pi(X)\,\mathrm{LR}(X)+c}{\Pi(X)+c}. \tag{5}$$

Thus along the cash line the loss ratio is the combination of the loss ratio of the book and the loss ratio 1 of the sure loss, with the relative costs of the two as weights: adding a sure loss pulls the ratio towards 1, and the singularity occurs where the relative cost of the book becomes infinite, that is, at $c = -\Pi(X)$. Proposition 4.13 below shows that this makes every cash line a unit-slope Möbius map.

If $\Pi(X)$ and $\Pi(Y)$ have the same sign, then $w_{X,Y} \in [0,1]$ and (CM) makes $\mathrm{LR}(X+Y)$ a convex combination of $\mathrm{LR}(X)$ and $\mathrm{LR}(Y)$; in particular it implies the mediant inequality: $\mathrm{LR}(X+Y)$ lies between $\mathrm{LR}(X)$ and $\mathrm{LR}(Y)$. The inequality is a betweenness property, and, as in decision theory, where betweenness yields only implicit representations while weighted averages yield ratios of linear functionals (Chew, 1983; Dekel, 1986), it does not suffice for our purposes (Example 9.3).

Given (P) and (N), (CM) holds if and only if the implied premium is comonotonic additive and $\Pi \cdot \mathrm{LR}$, restricted to D, extends to a comonotonic additive functional (Lemma 7.5).

Remark 4.11 (A premium given from outside). Suppose that a premium functional Π° is given and (CM) is stated with Π° as weights. If $\mathrm{LR}(X) = a \neq b = \mathrm{LR}(Y)$ and $\mathrm{LR}(X+Y) = L$, then $\Pi^\circ(X)/\Pi^\circ(Y) = (b-L)/(L-a)$, so the weights are determined by the loss ratios. On books with $\mathrm{LR} = 1$, however, the pooled ratio equals 1 whatever the weights, and (CM) carries no information on Π°. For instance, if $\mathrm{LR} \equiv 1$ on $\{I_g \neq 0\}$, then (CM) holds with the non-additive weights $\Pi^\circ = (I_g)^3$, since $a^3 + b^3 \neq 0$ implies $a + b \neq 0$. With the implied premium this difficulty does not arise, because (P) locates the price point also when the ratio equals 1 elsewhere on the cash line.

4.5 The implied risk measure

Definition 4.12 (Implied risk measure)**.** Under (P), put $\rho(X) := \Pi(X)\,\mathrm{LR}(X)$ for $X \in D$.

Proposition 4.13. *Assume* (P), (N) *and* (CM). *Then* ρ *extends uniquely to a cash-additive functional on* L^∞, *with* $\rho(c) = c$, *and for every* $X \in L^\infty$:

(a) $\mathrm{LR}(X+c) = \dfrac{\rho(X)+c}{\Pi(X)+c}$ *for every* $c \neq -\Pi(X)$;
(b) $\rho(X) = (\Pi(X)+c)\,\mathrm{LR}(X+c) - c$ *for every* $c \neq -\Pi(X)$;
(c) $\rho(X) - \Pi(X) = \lim_{c \to -\Pi(X)} (c+\Pi(X))\,\mathrm{LR}(X+c)$; *in particular* $\rho(X) = \lim_{c\to 0} c\,\mathrm{LR}(X+c)$ *if* $\Pi(X) = 0$;
(d) if $\rho(X) \neq \Pi(X)$, *then* $-\rho(X)$ *is the unique zero of the cash line of* X; *if* $\rho(X) = \Pi(X)$, *the cash line equals* 1 *away from the price point.*

The proof is given in Section 7. By (c), the implied risk measure of a book with zero premium is the residue of its cash line at the price point: if a small premium c is introduced, $\mathrm{LR}(X+c) \approx \rho(X)/c$. We call $N := \{\rho = \Pi\}$ the *break-even set*; $N \cap D = \{\mathrm{LR} = 1\}$, and N also contains the books with $\rho = \Pi = 0$. On N the residue vanishes and, by (d), the cash line has no zero, so that the risk measure cannot be read from the zero of the cash line; this is why it is defined through (b).

Corollary 4.14 (The implied risk measure is minus the zero)**.** *Assume* (P), (N) *and* (CM). *Then every cash line has at most one zero, and* $\rho(X) = -c_X^0$ *for every* $X \in L^\infty$.

Proof. By Proposition 4.13(d), if $\rho(X) \neq \Pi(X)$ the cash line of X has the unique zero $-\rho(X)$. If $\rho(X) = \Pi(X)$, it equals 1 away from the price point and has no zero, so $c_X^0 = c_X^\infty = -\Pi(X) = -\rho(X)$. □

Thus the implied risk measure, like the implied premium, is a point of the cash line: $\rho(X)$ is minus the sure amount at which the loss ratio of X vanishes, and on the break-even set the convention of Definition 4.7 makes the two points coincide.

Away from the break-even set, the premium can be computed explicitly from two values of the loss ratio. Solving (5) for $\Pi(X)$ gives the following formulas.

Proposition 4.15 (Explicit formulas)**.** *Assume* (P), (N) *and* (CM), *and let* $X \in D$ *with* $\mathrm{LR}(X) \neq 1$. *Write* $\phi_X(c) = \mathrm{LR}(X+c)$.

(a) Two quotes. *For every* $c \notin \{0, -\Pi(X)\}$, $\phi_X(c) \neq \mathrm{LR}(X)$ *and*

$$\Pi(X) = \frac{c\,\big(1-\phi_X(c)\big)}{\phi_X(c) - \mathrm{LR}(X)}, \qquad \rho(X) = \mathrm{LR}(X)\,\frac{c\,\big(1-\phi_X(c)\big)}{\phi_X(c) - \mathrm{LR}(X)}.$$

(b) Sensitivity to a sure loss. ϕ_X *is differentiable at* 0, $\phi_X'(0) = (1-\mathrm{LR}(X))/\Pi(X) \neq 0$, *and*

$$\Pi(X) = \frac{1-\mathrm{LR}(X)}{\phi_X'(0)}, \qquad \rho(X) = \frac{\mathrm{LR}(X)\,\big(1-\mathrm{LR}(X)\big)}{\phi_X'(0)}.$$

(c) Underwriting result. $\Pi(X) - \rho(X) = \lim_{|c|\to\infty} c\,\big(1-\phi_X(c)\big)$.

If $\mathrm{LR}(X) = 1$, *then* $\phi_X \equiv 1$ *away from the price point, and* $\Pi(X)$ *is not determined by the finite values of the cash line.*

Proof. By Proposition 4.13(a), $\phi_X(c) = (\rho+c)/(\Pi+c)$ with $\rho = \rho(X)$, $\Pi = \Pi(X) \neq 0$, so

$$1 - \phi_X(c) = \frac{\Pi-\rho}{\Pi+c}, \qquad \phi_X(c) - \mathrm{LR}(X) = \frac{c\,(\Pi-\rho)}{\Pi\,(\Pi+c)}.$$

Since $\mathrm{LR}(X) \neq 1$ means $\rho \neq \Pi$, the second expression is non-zero for $c \neq 0$, and the quotient gives (a). Differentiating, $\phi_X'(c) = (\Pi-\rho)/(\Pi+c)^2$, so $\phi_X'(0) = (1-\mathrm{LR}(X))/\Pi$, which gives (b); and $c\,(1-\phi_X(c)) = c\,(\Pi-\rho)/(\Pi+c) \to \Pi-\rho$, which gives (c). If $\mathrm{LR}(X) = 1$, then $\rho = \Pi$ and $\phi_X(c) = 1$ for $c \neq -\Pi$. □

Formula (b) reads: the implied premium is the underwriting margin $1-\mathrm{LR}(X)$ divided by the sensitivity of the loss ratio to a sure loss; formula (a) is its finite-difference version, which needs only two quoted loss ratios, with and without a sure loss c. For $X=\mathbf{1}_A$ in Example 4.17, $\mathrm{LR}(X)=\frac{1}{2}$ and $\phi_X(1)=\frac{5}{6}$, so (a) gives $\Pi(X)=\frac{1}{6}/\frac{1}{3}=\frac{1}{2}$. For books outside D the formulas apply to $X+c$ with $X+c\in D$, and $\Pi(X)=\Pi(X+c)-c$ by (4).

The last statement of Proposition 4.15 is the reason why the axioms must refer to non-finite values of LR, as ($\mathsf{P}_\exists$) and (R) do. On the break-even set the cash line carries no information about the premium except the location of its singularity, and Remark 4.11 shows that the weights in (CM) are then not identified by the values of the ratio. Off the break-even set, the weights of (CM) could equally be written with the explicit premium of (a) or (b) in place of $-c^\infty$; but (CM) must also govern pools that contain break-even books, and there only the price point locates the premium. Axiom (R) avoids weights altogether: it moves each book to zero premium by a sure payment, which is a statement about non-finite values, and pools there.

Remark 4.16 (A weight-free form of (CM)). Property (CM) is a cardinal statement whose weights are read off the price points. Given (P), the pair (N)+(CM) is equivalent to the following two statements, which use no weights:

(M) every cash line has at most one zero, and $\mathrm{LR}(X+c)=\dfrac{c-c_X^0}{c-c_X^\infty}$ for all $c\neq c_X^\infty$;

(A) if X,Y are comonotone, then $c_{X+Y}^\infty=c_X^\infty+c_Y^\infty$ and $c_{X+Y}^0=c_X^0+c_Y^0$.

(M) says that every cash line is the unit-slope Möbius map determined by its own zero and price point; (A) says that comonotone pooling adds the cash offsets at the levels 0 and ∞. Indeed, (N) and (CM) imply (M) by Proposition 4.13(a) and Corollary 4.14, and (A) by Lemma 7.5(a). Conversely, (A) with $X=Y=0$ gives $c_0^\infty=c_0^0=0$, so (M) gives $\mathrm{LR}(c)=c/c=1$ for $c\neq 0$, which is (N). If $X,Y\in D$ are comonotone with $c_X^\infty+c_Y^\infty\neq 0$, then $X+Y\in D$ by (A), and (M) at $c=0$ gives $\mathrm{LR}(Z)=c_Z^0/c_Z^\infty$ for $Z\in\{X,Y,X+Y\}$; by (A), $\mathrm{LR}(X+Y)=(c_X^0+c_Y^0)/(c_X^\infty+c_Y^\infty)=w_{X,Y}\mathrm{LR}(X)+w_{Y,X}\mathrm{LR}(Y)$, which is (CM). Unlike the formulation (CP) of Remark 4.19, (M) refers to no functional other than LR, but it still uses the points c_X^∞ and c_X^0. Axiom (R) states the content of (M) and (A) without them.

Example 4.17 (Three cash lines)**.** Let $\rho=I_{\mathrm{id}}=\mathbb{E}$ and $\Pi=I_{\sqrt{\cdot}}$, and let $P(A)=\frac{1}{4}$. Figure 1 shows the cash lines of three books.

(i) $X=\mathbf{1}_A$: $\mathrm{LR}(X+c)=(\frac{1}{4}+c)/(\frac{1}{2}+c)$. The price point is $-\frac{1}{2}$, so $\Pi(X)=\frac{1}{2}$, and $\rho(X)=\Pi(X)\mathrm{LR}(X)=\frac{1}{4}$; the cash line vanishes at $-\frac{1}{4}$.

(ii) $Y=\mathbf{1}_A-\frac{1}{2}$, a book with zero premium: $\mathrm{LR}(Y)=\infty$ and $\mathrm{LR}(Y+c)=(c-\frac{1}{4})/c$. The residue at the price point 0 is $-\frac{1}{4}=\rho(Y)$.

(iii) $Z\equiv\frac{1}{2}$, a break-even book: $\mathrm{LR}(Z+c)=1$ for $c\neq-\frac{1}{2}$, and $\mathrm{LR}(Z-\frac{1}{2})=\star$. The price point gives $\Pi(Z)=\frac{1}{2}$; the cash line has no zero, and $\rho(Z)=\Pi(Z)\cdot 1=\frac{1}{2}$.

Remark 4.18 (Labels at the price point). To have $\mathrm{LR}=[\rho:\Pi]$ also at price points, one may adopt the convention

(S) at the price point, $\mathrm{LR}=\star$ if the residue in (c) is zero and $\mathrm{LR}=\infty$ otherwise.

Convention (S) is not used in the proofs. If, instead, the loss ratio is real valued with $0/0:=1$, the price point of a break-even book cannot be observed and the implied premium is not defined there.

Remark 4.19 (A formulation by cash lines). One may alternatively postulate (CP): every cash line is a unit-slope Möbius map, $\mathrm{LR}(X+c)=[\rho(X)+c:\Pi(X)+c]$. Proposition 4.13(a) shows that (CP) follows from (P), (N) and (CM), the labels agreeing under (S). Conversely, (CP) implies (P) and (S), and (CP) together with (CM) implies (N): if $\pi_0:=\Pi(0)\neq 0$, (CM) applied to the constants $a=b=-\pi_0/2$ forces $\mathrm{LR}(-\pi_0)\in\mathbb{R}$, although $-\pi_0$ is the price point of 0; hence $\pi_0=0$, and (CM) with $X=Y=a\neq 0$ gives $(\rho(0)+2a)/(2a)=(\rho(0)+a)/a$, i.e. $\rho(0)=0$ and $\mathrm{LR}(c)=1$. Thus (P)+(N)+(CM)+(S) is equivalent to (CP)+(CM).

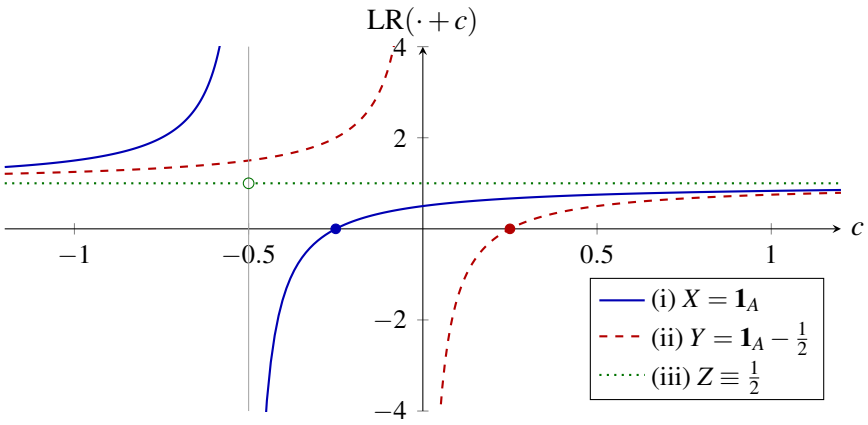


Figure 1: Cash lines of the three books of Example 4.17. Book (i) explodes at $c = -\frac{1}{2}$ (its price point) and vanishes at $c = -\frac{1}{4}$; book (ii) explodes at $c = 0$ with residue $-\frac{1}{4}$; book (iii) equals 1 except at its price point $c = -\frac{1}{2}$, where the singularity is removable (open circle).

4.6 Axiom (LI): law invariance

Axiom 4.20 ((LI))**.** If $X \overset{d}{=} Y$, then either $\mathrm{LR}(X) = \mathrm{LR}(Y)$ or both values are non-finite.

This is the standard assumption of premium and risk-measure theory (Wang et al., 1997; Kusuoka, 2001). Because the implied premium and risk measure are computed from cash lines, (LI) passes to them (Lemma 7.6). Ratios that weight scenarios non-uniformly, such as $\mathbb{E}_Q/\mathbb{E}$ with $Q \neq P$, violate (LI); they are covered by Theorem 5.7.

4.7 Axioms (RC) and (RM): regularity of the cash offsets

The regularity axioms concern two sure amounts attached to a book: the amount a whose payment leaves the book unpriced, $\mathrm{LR}(X-a) \notin \mathbb{R}$, and the amount a whose payment leaves it without loss. A book W has no loss if $\mathrm{LR}(W) = 0$, or, when W is unpriced, if $\mathrm{LR}(W+1) = 1$, that is, if W written at the premium 1 costs exactly that premium.

Axiom 4.21 ((RC))**.** For every $\varepsilon > 0$ there is $\delta > 0$ such that for all $X, Y \in L^\infty$ with $\|X - Y\|_\infty < \delta$ and all $a, b \in \mathbb{R}$:

(i) if $\mathrm{LR}(X-a) \notin \mathbb{R}$ and $\mathrm{LR}(Y-b) \notin \mathbb{R}$, then $|a-b| < \varepsilon$;
(ii) if each $W \in \{X-a, Y-b\}$ satisfies $[\mathrm{LR}(W) = 0$, or $\mathrm{LR}(W) \notin \mathbb{R}$ and $\mathrm{LR}(W+1) = 1]$, then $|a-b| < \varepsilon$.

Axiom 4.22 ((RM))**.** Let $X \leq Y$ almost surely and $a, b \in \mathbb{R}$.

(i) If $\mathrm{LR}(X-a) \notin \mathbb{R}$ and $\mathrm{LR}(Y-b) \notin \mathbb{R}$, then $a \leq b$.
(ii) If each $W \in \{X-a, Y-b\}$ satisfies $[\mathrm{LR}(W) = 0$, or $\mathrm{LR}(W) \notin \mathbb{R}$ and $\mathrm{LR}(W+1) = 1]$, then $a \leq b$.

Taking $X = Y$ shows that each axiom forces the two amounts to be unique, so the axioms make sense independently of the others. Under ($\mathsf{P}_\exists$) and (R), the amount in (i) is the implied premium $\Pi(X)$ and the amount in (ii) is the implied risk measure $\rho(X)$ (Lemma 7.2); thus (RC) (respectively (RM)) says exactly that Π and ρ are uniformly continuous (monotone) on L^∞, and (RM) implies (RC) (Lemma 7.7). In terms of the cash lines, the two amounts are minus the price point and minus the zero. Axiom (RM) says that a larger loss must be offset by a larger sure amount, both to be priced at zero (level ∞) and to have no loss (level 0). This is the monotonicity of the cash offsets familiar from acceptability indices (Cherny and Madan, 2009), imposed at the two levels that determine the cash line; (RC) is the

corresponding continuity. For a ratio $[A:B]$ with A, B cash additive, the two amounts are $B(X)$ and $A(X)$, so that the axioms reduce to continuity or monotonicity of A and B. We include the signed tier (RC) because the canonical family ρ_β is a difference of monotone functionals and is in general not monotone (Corollary 10.1 and Example 10.2). Monotonicity is imposed on the cash offsets and not on the values of the ratio, for the following reason.

Proposition 4.23 (No monotone loss ratio)**.** *Let $h, g \in \mathscr{H}_1$ with $g > 0$ on $(0,1]$, and let $\mathrm{LR} = I_h/I_g$ on $\mathscr{X} := L^\infty_+ \setminus \{0\}$. If LR is monotone on $\mathscr{X}$ (non-decreasing or non-increasing for the almost sure order), then $h = g$ and $\mathrm{LR} \equiv 1$.*

Proof. Let $P(A) = p > 0$. For $c \geq 0$, $\mathrm{LR}(\mathbf{1}_A + c) = \frac{h(p)+c}{g(p)+c}$ is monotone in c with derivative at 0 of the sign of $g(p) - h(p)$. Next let $P(E) = q > 0$ and $t \geq 0$. Then $X_t = 1 + t\mathbf{1}_E \in \mathscr{X}$ is increasing in t, and $I_k(X_t) = 1 + t\,k(q)$ for $k \in \{h, g\}$, so $\mathrm{LR}(X_t) = \frac{1+t\,h(q)}{1+t\,g(q)}$ has derivative at $t = 0$ of the sign of $h(q) - g(q)$. If LR is non-decreasing both signs are non-negative, hence $h = g$ on $(0,1]$; if LR is non-increasing both are non-positive, and again $h = g$. □

Example 4.24. Let $h = \mathrm{id}$, $g = \sqrt{\cdot}$ and $p = q = \frac{1}{4}$ in the proof. Both families of books in Figure 2 increase in the almost sure order, but the loss ratio increases along the first and decreases along the second: adding a sure loss to an over-priced book moves the ratio towards 1 from below, whereas adding risk to a sure loss moves it away from 1.

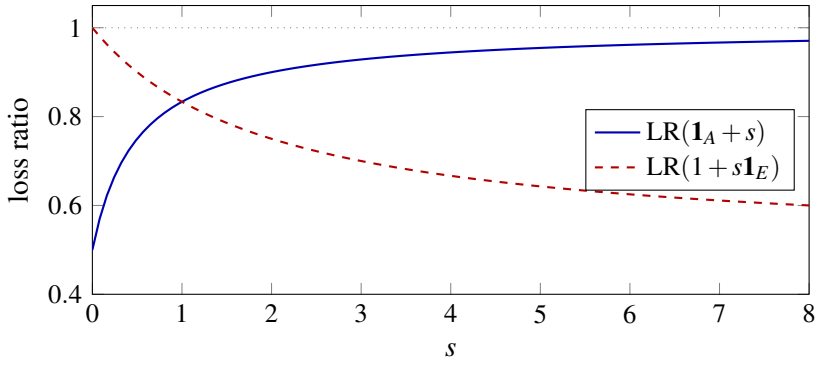


Figure 2: Example 4.24: $h = \mathrm{id}$, $g = \sqrt{\cdot}$, $P(A) = P(E) = \frac{1}{4}$. Both families of books increase with s; the loss ratio increases along one and decreases along the other.

Consequently, the theorems on acceptability indices (Cherny and Madan, 2009), which require monotonicity of the index, do not apply to loss ratios. Axiom (RM) is nevertheless an order property of LR: it concerns the sure amounts at which the ratio becomes non-finite or vanishes, not the size of the ratio.

4.8 Summary

5 Main results

Throughout this section $\mathrm{LR} : L^\infty \to \widehat{\mathbb{R}}_\star$, and Π and ρ denote the implied premium and the implied risk measure (Definitions 4.8 and 4.12, Proposition 4.13).

5.1 The two characterization theorems

Theorem 5.1 (Signed tier)**.** *The following are equivalent.*

Axiom	Statement	Interpretation and counterpart
($\mathsf{P}_\exists$)	each cash line has a non-finite value	every book can be made unpriced by a sure payment
(R)	comonotone pooling of unpriced books, (1)	underwriting results add; comonotonic additivity (Yaari, 1987; Wang et al., 1997)
(LI)	law invariance	standard (Wang et al., 1997; Kusuoka, 2001)
(RC)/(RM)	regularity of the sure amounts that make a book unpriced and loss-free	continuity / monotonicity of premium and risk measure; cash offsets (Cherny and Madan, 2009)
Derived properties, equivalent to ($\mathsf{P}_\exists$) *and* (R) *(Proposition 4.5)*		
(P)	one price point on each cash line	defines the premium; the level $\beta = \infty$ of the associated family
(N)	$\mathrm{LR}(c) = 1$ for $c \neq 0$	no unjustified loading (Gerber, 1974)
(CM)	comonotone pooling, weighted by relative price read off the price points	premium-weighted average

Table 2: The axioms and the derived properties. Every axiom refers only to values of LR.

(i) LR *satisfies* ($\mathsf{P}_\exists$), (R), (LI) *and* (RC).
(ii) *There is a pair* $(h,g) \in \mathscr{H}_1^2$ *such that* $\mathrm{LR}(X) = I_h(X)/I_g(X)$ *whenever* $I_g(X) \neq 0$, *and* $\mathrm{LR}(X) \notin \mathbb{R}$ *whenever* $I_g(X) = 0$.

In that case the price point and the zero of the cash line of X *are* $-I_g(X)$ *and* $-I_h(X)$*; that is, the implied premium is* $\Pi = I_g$ *and the implied risk measure is* $\rho = I_h$. *If, in addition, the labelling convention* (S) *holds, then* $\mathrm{LR} = [I_h : I_g]$ *on all of* L^∞.

Theorem 5.2 (Distortion tier)**.** *The following are equivalent.*

(i) LR *satisfies* ($\mathsf{P}_\exists$), (R), (LI) *and* (RM).
(ii) *Statement (ii) of Theorem 5.1 holds with* h, g *distortion functions (non-decreasing elements of* $\mathscr{H}_1$*).*

By Proposition 4.5, condition (i) may equivalently be stated with the properties (P), (N) and (CM) in place of the axioms ($\mathsf{P}_\exists$) and (R). The proofs are given in Section 7; Section 9 shows that none of the axioms can be dropped.

5.2 Uniqueness and derived properties

Proposition 5.3. *In Theorems 5.1 and 5.2,* (h,g) *is unique. For an event* A *with* $P(A) = p$,

$$g(p) = -c^{\infty}_{\mathbf{1}_A}, \qquad c^{\infty}_{\mathbf{1}_A} = \text{the price point of } \mathbf{1}_A;$$
$$h(p) = \big(g(p) + c\big)\,\mathrm{LR}(\mathbf{1}_A + c) - c \qquad \text{for any } c \neq -g(p).$$

Moreover $D = \{I_g \neq 0\}$ *and the break-even set is* $N := \{\rho = \Pi\} = \{I_{h-g} = 0\}$. *Within Theorem 5.1,* ρ *(respectively* Π*) is monotone if and only if* h *(respectively* g*) is non-decreasing.*

Proposition 5.4. *Under the axioms of Theorem 5.1:*

(i) *every cash line is a unit-slope Möbius map,* $\mathrm{LR}(X + c) = (\rho(X) + c)/(\Pi(X) + c)$ *for* $c \neq -\Pi(X)$;
(ii) ρ *and* Π *are cash additive,* $\rho(c) = \Pi(c) = c$, *and they are Lipschitz with constants* $\mathrm{TV}(h)$ *and* $\mathrm{TV}(g)$;
(iii) ρ *and* Π *are comonotonic additive and positively homogeneous, hence* LR *is scale invariant:* $\mathrm{LR}(\lambda X) = \mathrm{LR}(X)$ *for* $\lambda > 0$ *and* $X \in D$, *and for all* X *under* (S).

Properties that are often postulated for the numerator and the denominator of a loss ratio are thus derived; the association itself is derived in Corollary 10.1.

5.3 Examples

Example 5.5 (Expected loss over a proportional hazard premium)**.** Let $h = \mathrm{id}$ and $g_r(p) = p^r$ with $r \in (0,1)$, so that $\rho = \mathbb{E}$ and Π is the proportional hazard premium. For $P(A) = p$, $\mathrm{LR}(\mathbf{1}_A) = p^{1-r}$, and Proposition 5.3 recovers the premium distortion from the loss ratios of Bernoulli books, $g_r(p) = p/\mathrm{LR}(\mathbf{1}_A)$. For a loss U uniform on $[0,1]$, $\mathbb{E}U = \frac{1}{2}$ and $\Pi(U) = \int_0^1 (1-x)^r\,dx = 1/(1+r)$, so $\mathrm{LR}(U) = (1+r)/2$. Since g_r is concave and h is linear, $\mathrm{LR} \le 1$ on $\mathscr{X}$ (Corollary 10.4). The left panel of Figure 3 shows $p \mapsto \mathrm{LR}(\mathbf{1}_A)$ for three values of r.

Example 5.6 (Expected shortfall over expected loss)**.** Let $h_\alpha(p) = \min\{p/(1-\alpha), 1\}$, $\alpha \in (0,1)$, and $g = \mathrm{id}$, so that $\rho = \mathrm{ES}_\alpha$ and $\Pi = \mathbb{E}$. Then $\mathrm{LR}(\mathbf{1}_A) = \min\{1/(1-\alpha), 1/p\}$ and $\mathrm{LR}(U) = 1+\alpha$ for U uniform on $[0,1]$. Here h_α is concave and g is convex, so $\mathrm{LR} \ge 1$ and LR is quasi-convex on $\{\Pi > 0\}$ (Corollary 10.4(d)); see the right panel of Figure 3.

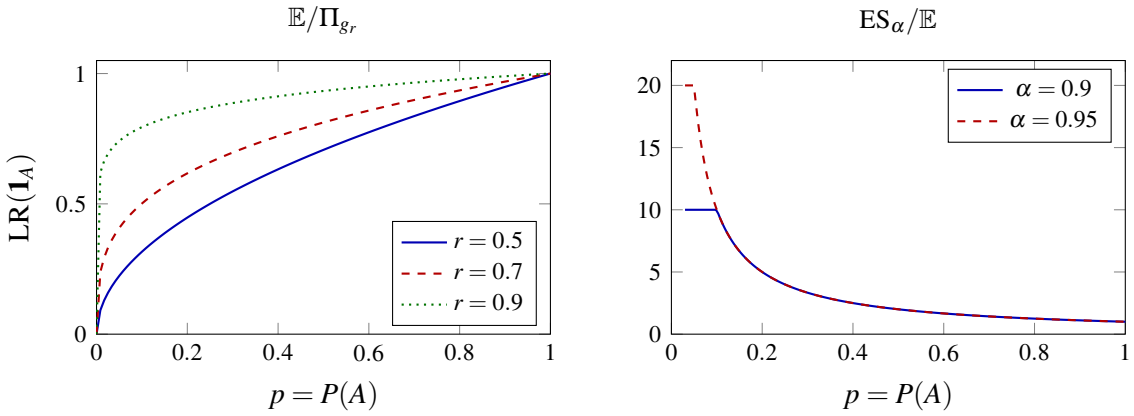


Figure 3: Loss ratios of Bernoulli books $\mathbf{1}_A$ as functions of $p = P(A)$. Left: expected loss over the proportional hazard premium (Example 5.5); the curves identify $g_r(p) = p/\mathrm{LR}(\mathbf{1}_A)$. Right: expected shortfall over expected loss (Example 5.6).

5.4 Dropping law invariance

We state the version without (LI), which covers scenario-dependent loss ratios. A *game* is a map $\nu : \mathscr{F} \to \mathbb{R}$ with $\nu(\emptyset) = 0$. It is *null-invariant* if $P(A \triangle B) = 0$ implies $\nu(A) = \nu(B)$; its *chain variation* is $\|\nu\|_{bv} = \sup \sum_j |\nu(A_j) - \nu(A_{j-1})|$ over chains $\emptyset = A_0 \subset \cdots \subset A_n = \Omega$. Let $\mathscr{V}_1$ be the null-invariant games with $\|\nu\|_{bv} < \infty$ and $\nu(\Omega) = 1$, and for $\nu \in \mathscr{V}_1$ let $C_\nu(X) = \int_0^\infty \nu(X > x)\,dx + \int_{-\infty}^0 [\nu(X > x) - 1]\,dx$. A *capacity* is a monotone game.

Theorem 5.7 (Without law invariance)**.** LR *satisfies* ($\mathsf{P}_\exists$), (R) *and* (RC) *if and only if* $\mathrm{LR} = C_{\nu_1}/C_{\nu_2}$ *where* $C_{\nu_2} \neq 0$ *and* $\mathrm{LR} \notin \mathbb{R}$ *where* $C_{\nu_2} = 0$*, for* $\nu_1, \nu_2 \in \mathscr{V}_1$*; with* (RM) *in place of* (RC)*, if and only if the same holds with capacities* ν_1, ν_2*. The pair is unique:* $\nu_1(A) = \rho(\mathbf{1}_A)$ *and* $\nu_2(A) = \Pi(\mathbf{1}_A)$*. The loss ratio is in addition law invariant if and only if* $\nu_i = k_i \circ P$ *with* $k_i \in \mathscr{H}_1$*, which recovers Theorems 5.1 and 5.2.*

The set-function representation in Theorem 5.7 is that of Murofushi et al. (1994) and Cerreia-Vioglio et al. (2012, 2015); the proof is in Appendix A.

6 On the role of the axioms

6.1 Structure of the result

The proof of Theorem 5.1 has three steps.

(A) *Implied pair.* Axioms ($\mathsf{P}_\exists$) and (R) give (P), (N) and (CM) (Lemma 7.1). Property (P) defines Π; (R) with a sure loss, equivalently (N) and (CM) applied with sure losses, defines ρ and gives the Möbius form of the cash lines (Lemmas 7.1, 7.3 and 7.4).

(B) *Transfer.* Given (P) and (N), (CM) holds if and only if ρ and Π are comonotonic additive (Lemma 7.5), and (LI) passes from LR to ρ and Π (Lemma 7.6).

(C) *Representation.* A law-invariant, comonotonic additive, uniformly continuous functional on an atomless L^∞ is a signed Choquet integral (Wang et al., 2020a, Thm. 1); under monotonicity it is a distortion premium (Schmeidler, 1986; Wang et al., 1997).

Step (C) is classical; we reprove it in Section 7 to keep the constants explicit and to show that no continuity is needed in the monotone case. Steps (A) and (B) are specific to loss ratios. Together, axioms ($\mathsf{P}_\exists$) and (R) state that LR is the ratio of a pair of comonotonic additive functionals that coincide with the identity on constants, and that this pair can be read off LR.

6.2 Assumed and derived properties

Table 3 lists the properties of the representation and their source. Axiom ($\mathsf{P}_\exists$) is ordinal, and (LI), (RC), (RM) select the class of representing functions. The cardinal content of the axioms lies in (R), which contains the normalization (N) and the pooling property (CM); given (P) and (N), (CM) is equivalent to comonotonic additivity of the implied premium and risk measure, which is the axiom that produces Choquet representations in Yaari (1987), Schmeidler (1986) and Wang et al. (1997). The Choquet form of the numerator and the denominator is therefore a consequence of comonotonic additivity, as in the single-functional case.

Property	**Source**	**Status**
unique price point ((P)); Π cash additive	($\mathsf{P}_\exists$) and (R) (Lemma 7.1)	derived
$\mathrm{LR}(c)=1$ ((N)); $\Pi(c)=c$, $\rho(c)=c$	(R) (Lemma 7.1)	derived
ρ defined on L^∞; Möbius cash lines	(R) with a sure loss (Lemma 7.1)	derived
ρ,Π comonotonic additive; (CM)	(R) (Lemmas 7.1 and 7.5)	equivalent
ρ,Π law invariant	(LI) (Lemma 7.6)	assumed
ρ,Π continuous or monotone	(RC), (RM)	assumed
h,g of bounded variation	Lemma 7.9	derived
$\rho=I_h$, $\Pi=I_g$	Lemmas 7.8 and 7.10	derived
positive homogeneity, scale invariance	Lemma 7.7	derived
association	Corollary 10.1	derived
uniqueness of (h,g)	Proposition 5.3	derived

Table 3: Assumed and derived properties in Theorem 5.1.

6.3 Relation to the single-functional theory

The characterizations in the literature concern premiums and risk measures; they cannot be applied to LR directly, since a loss ratio is not additive and (R) is an identity between loss ratios. What is specific to the loss ratio is the construction of the implied pair, the derivation of additivity on all comonotone pairs, in particular on the break-even set where the weights of (CM) are not identified (Remark 4.11), uniqueness of (h,g) also when $\mathrm{LR}\equiv 1$, the impossibility of a monotone loss ratio (Proposition 4.23), the structure of the associated family (Corollary 10.1), and the version for nonnegative losses (Section 8), in which the price point is not observed and must be extrapolated.

6.4 Loss ratios excluded and included

Ratios with a proportional loading, $\mathbb{E}/((1+\theta)\mathbb{E})$, violate (R) through (N); scenario-weighted ratios $\mathbb{E}_Q/\mathbb{E}$ violate (LI); the ratio $\max(I_{\sqrt{t}}, I_{\min(1,1.5t)})/\mathbb{E}$ violates (R), its numerator being cash additive but not comonotonic additive; and ordinal transforms such as $(\rho/\Pi)^3$ satisfy ($\mathsf{P}_\exists$), (P) and (N) but violate (R). The class characterized contains, among others, the expected loss over any distortion premium (Example 5.5), ratios of expected shortfalls (Example 5.6), value-at-risk over expected shortfall, and ratios that take values on both sides of 1.

7 Proofs of the main results

Figure 4 shows the structure of the proof. Lemma 7.1 derives the properties (P), (N) and (CM) from the axioms ($\mathsf{P}_\exists$) and (R); Lemmas 7.1–7.6 are specific to loss ratios; Lemmas 7.7–7.10 reprove the representation of comonotonic additive functionals in the form needed here.

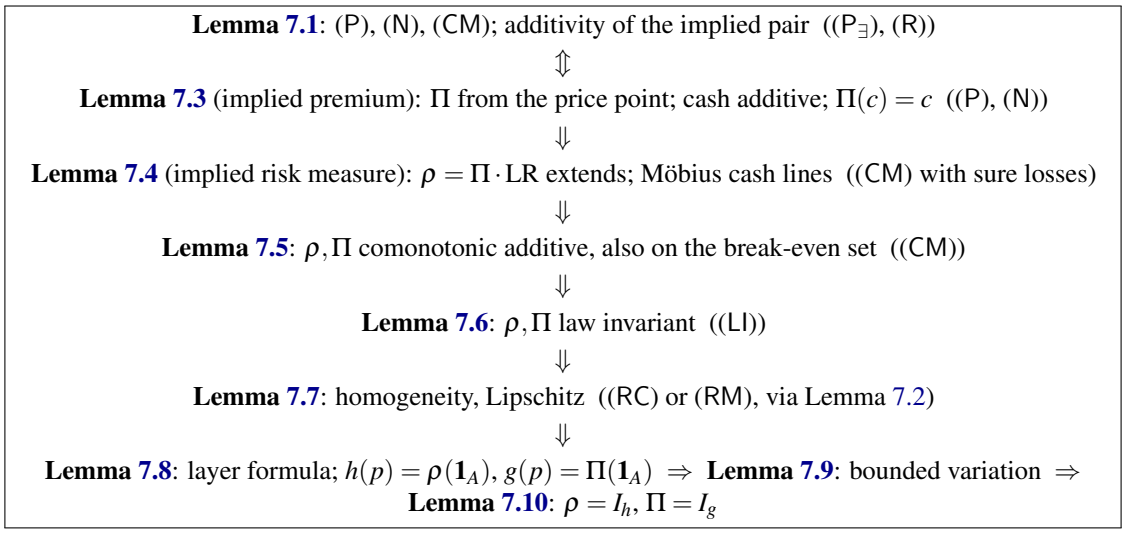


Figure 4: Logical structure of the proof of Theorems 5.1 and 5.2.

Lemma 7.1 (The derived properties)**.** *Axioms* ($\mathsf{P}_\exists$) *and* (R) *hold if and only if* (P)*,* (N) *and* (CM) *hold. In that case, for every unpriced W (that is, $\mathrm{LR}(W)\notin\mathbb{R}$) the quantity $r(W):=s\,(\mathrm{LR}(W+s)-1)$ does not depend on $s\neq 0$, and the implied premium and the implied risk measure satisfy $\Pi(X)=-c_X^\infty$ and $\rho(X)=\Pi(X)+r(X-\Pi(X))$ and are comonotonic additive.*

Proof. Assume ($\mathsf{P}_\exists$) and (R). The quantities Π,ρ,r below are used only in the proof; they appear in no axiom.

Step 0: (P). Let $\mathrm{LR}(Z) \notin \mathbb{R}$. The book Z is comonotone with itself, so (R) applied to the pair (Z,Z) gives $\mathrm{LR}(Z+s) \in \mathbb{R}$ for every $s \neq 0$. Hence a cash line has at most one non-finite point, and by ($\mathsf{P}_\exists$) exactly one. So (P) holds; put $\Pi(X) := -c_X^\infty$, so that $\Pi(X+c) = \Pi(X) + c$ by (3).

Step 1: (N). Let p_0 be the non-finite point of the cash line of 0. Constants are comonotone, and $\mathrm{LR}(p_0) \notin \mathbb{R}$, so (R) applied to (p_0, p_0) gives $\mathrm{LR}(2p_0 + u) \in \mathbb{R}$ for all $u \neq 0$. If $p_0 \neq 0$, the choice $u = -p_0$ gives $\mathrm{LR}(p_0) \in \mathbb{R}$, a contradiction; hence $p_0 = 0$ and $\mathrm{LR}(0) \notin \mathbb{R}$. Now apply (R) to $(0,0)$ with $s = t = u$: $f(u) := u(\mathrm{LR}(u) - 1)$ satisfies $f(u) = 2f(u)$, so $f(u) = 0$ and $\mathrm{LR}(u) = 1$ for every $u \neq 0$.

Step 2: the cash-line law. Let W be unpriced. The pair $(W,0)$ is comonotone and both books are unpriced, so (1) gives $u(\mathrm{LR}(W+u) - 1) = s(\mathrm{LR}(W+s) - 1) + t(\mathrm{LR}(t) - 1) = s(\mathrm{LR}(W+s) - 1)$ for all $s,t,u \neq 0$, by Step 1. Hence $r(W)$ does not depend on s. For any X put $X_0 := X - \Pi(X)$, which is unpriced, and $\rho(X) := \Pi(X) + r(X_0)$. For $c \neq -\Pi(X)$ write $X + c = X_0 + s$ with $s = \Pi(X) + c \neq 0$; then

$$\mathrm{LR}(X+c) = 1 + \frac{r(X_0)}{s} = \frac{\rho(X) + c}{\Pi(X) + c}.$$

In particular $\rho = \Pi \cdot \mathrm{LR}$ on D, ρ is cash additive, and $\rho(c) = \Pi(c) = c$.

Step 3: comonotonic additivity. Let X, Y be comonotone. Cash shifts preserve comonotonicity, so X_0 and Y_0 are comonotone and unpriced. By (R), $\mathrm{LR}(X_0 + Y_0 + u) \in \mathbb{R}$ for every $u \neq 0$; by (P) the non-finite point of the cash line of $X_0 + Y_0$ is therefore 0, so $\Pi(X_0 + Y_0) = 0$, that is, $\Pi(X+Y) = \Pi(X) + \Pi(Y)$. Consequently $(X+Y)_0 = X_0 + Y_0$, and (1) with $s = t = u$ gives $r(X_0 + Y_0) = r(X_0) + r(Y_0)$; hence $\rho(X+Y) = \rho(X) + \rho(Y)$. No premium is recovered from finite values, so the argument covers break-even books, pairs of which only one breaks even, the case $\mathrm{LR} \equiv 1$ on D, and pools whose premiums cancel. By Lemma 7.5(b), which needs only (P) and (N), (CM) holds, and the implied risk measure of Definition 4.12 is the functional ρ just constructed.

Converse. Assume (P), (N), (CM). Then ($\mathsf{P}_\exists$) is clear. By Proposition 4.13(a) and Lemma 7.5(a), every cash line is $\mathrm{LR}(X+c) = (\rho(X) + c)/(\Pi(X) + c)$ and ρ, Π are comonotonic additive. Let X, Y be comonotone and unpriced, so $\Pi(X) = \Pi(Y) = 0$. For $s \neq 0$, $\mathrm{LR}(X+s) = (\rho(X) + s)/s \in \mathbb{R}$ and $s(\mathrm{LR}(X+s) - 1) = \rho(X)$, and likewise for Y. Moreover $\Pi(X+Y) = 0$, so $\mathrm{LR}(X+Y+u) \in \mathbb{R}$ for $u \neq 0$ and $u(\mathrm{LR}(X+Y+u) - 1) = \rho(X+Y) = \rho(X) + \rho(Y)$, which is (1). □

Proof of Proposition 4.5. This is the first statement of Lemma 7.1. □

Lemma 7.2 (The cash offsets)**.** *Assume* ($\mathsf{P}_\exists$) *and* (R)*, and let $X \in L^\infty$, $a \in \mathbb{R}$ and $W = X - a$. Then* $\mathrm{LR}(W) \notin \mathbb{R}$ *if and only if* $a = \Pi(X)$*; and [*$\mathrm{LR}(W) = 0$*, or* $\mathrm{LR}(W) \notin \mathbb{R}$ *and* $\mathrm{LR}(W+1) = 1$*] if and only if* $a = \rho(X)$*. Consequently* (RC) *holds if and only if* Π *and* ρ *are uniformly continuous, and* (RM) *holds if and only if* Π *and* ρ *are monotone.*

Proof. The first statement is the definition of the price point. For the second, $\Pi(W) = \Pi(X) - a$ and $\rho(W) = \rho(X) - a$. If $\Pi(W) \neq 0$, then W is not unpriced and $\mathrm{LR}(W) = \rho(W)/\Pi(W)$, which vanishes exactly when $\rho(W) = 0$. If $\Pi(W) = 0$, then $\mathrm{LR}(W) \notin \mathbb{R}$ and $\mathrm{LR}(W+1) = \rho(W) + 1$ by Lemma 7.1, which equals 1 exactly when $\rho(W) = 0$. In both cases the condition means $a = \rho(X)$. The axioms (RC) and (RM) therefore compare $\Pi(X)$ with $\Pi(Y)$ in (i) and $\rho(X)$ with $\rho(Y)$ in (ii). □

Lemma 7.3 (Implied premium)**.** *Assume* (P)*. Then* $\Pi(X+c) = \Pi(X) + c$ *for all X and c, and* $D = \{\Pi \neq 0\}$*. If also* (N)*, then* $\Pi(c) = c$ *for every constant c.*

Proof. The cash line of $X + c$ is $d \mapsto \mathrm{LR}(X + c + d)$, the cash line of X shifted by c; its price point is therefore the price point of X minus c, which is (4). $X \in D$ means that 0 is not the price point, i.e. $\Pi(X) \neq 0$. Under (N) the cash line of 0 takes the value 1 at every $d \neq 0$, so its price point is 0, $\Pi(0) = 0$, and $\Pi(c) = c$. □

Lemma 7.4 (Implied risk measure)**.** *Assume* (P)*,* (N)*,* (CM)*. If $X \in D$ and $X+c \in D$, then $\rho(X+c) = \rho(X)+c$. Consequently $\rho(X) := \rho(X+c)-c$, for any $c \neq -\Pi(X)$, defines a cash-additive functional on L^∞ that agrees with $\Pi \cdot \mathrm{LR}$ on D, satisfies $\rho(c) = c$, and*

$$\mathrm{LR}(X+c) = \frac{\rho(X)+c}{\Pi(X)+c} \qquad \text{for all } c \neq -\Pi(X). \tag{6}$$

Proof. Let $X, X+c \in D$ with $c \neq 0$. The constant c lies in D (by (N)), is comonotone with X, and $\Pi(X)+\Pi(c) = \Pi(X+c) \neq 0$ by Lemma 7.3. So (CM) applies to the pair (X,c) and gives $\mathrm{LR}(X+c) = (\Pi(X)\mathrm{LR}(X)+c)/(\Pi(X)+c)$, i.e. $\rho(X+c) = \Pi(X+c)\mathrm{LR}(X+c) = \rho(X)+c$.

For arbitrary X, $X+c \in D$ exactly when $c \neq -\Pi(X)$. If c, c' are two such values, then $X+c' = (X+c)+(c'-c)$ with both $X+c$ and $X+c'$ in D, so $\rho(X+c') = \rho(X+c)+(c'-c)$; hence $\rho(X+c)-c$ does not depend on c, and for $X \in D$ (take $c=0$) it equals $\Pi(X)\mathrm{LR}(X)$. Cash additivity of the extension is immediate. For $c \neq -\Pi(X)$, $\mathrm{LR}(X+c) = \rho(X+c)/\Pi(X+c) = (\rho(X)+c)/(\Pi(X)+c)$. Finally $\rho(c) = \Pi(c)\mathrm{LR}(c) = c$ for $c \neq 0$, and $\rho(0) = \rho(1)-1 = 0$. □

Proof of Proposition 4.13. (a) is (6), and (b) restates it. For (c), $(c+\Pi(X))\mathrm{LR}(X+c) = \rho(X)+c \to \rho(X)-\Pi(X)$ as $c \to -\Pi(X)$. For (d), the right-hand side of (6) vanishes only at $c = -\rho(X)$, which differs from the price point when $\rho(X) \neq \Pi(X)$; when $\rho(X) = \Pi(X)$ it equals 1 for every $c \neq -\Pi(X)$. Uniqueness of the extension is clear, since any cash-additive extension must satisfy (b). □

Lemma 7.5 (Comonotonic additivity)**.** *Assume* (P) *and* (N)*.*

(a) If (CM) *holds, then ρ and Π are comonotonic additive.*
(b) Conversely, if Π is comonotonic additive and there is a comonotonic additive $\tilde{\rho}$ with $\tilde{\rho} = \Pi \cdot \mathrm{LR}$ on D, then (CM) *holds and the implied risk measure equals $\tilde{\rho}$.*

Proof. (a) Let X, Y be comonotone, $Z = X+Y$, $r = \rho(X)+\rho(Y)$ and $q = \Pi(X)+\Pi(Y)$. Fix $s \neq -q$ and choose $u \notin \{-\Pi(X), s+\Pi(Y)\}$.

(1) $X+u$ and $Y+s-u$ are comonotone, and by Lemma 7.3 their premiums $\Pi(X)+u$ and $\Pi(Y)+s-u$ are non-zero. So both lie in D, and their premiums sum to $q+s \neq 0$.
(2) Apply (CM) to this pair and use $\rho = \Pi \cdot \mathrm{LR}$ on D with Lemma 7.4:

$$\mathrm{LR}(Z+s) = \frac{(\rho(X)+u)+(\rho(Y)+s-u)}{q+s} = \frac{r+s}{q+s} \in \mathbb{R}.$$

(3) Thus the cash line of Z is finite at every $s \neq -q$. By (P) it has exactly one non-finite point, so that point is $-q$, and $\Pi(Z) = q$.
(4) For $s \neq -q$, Lemma 7.4 gives $\rho(Z)+s = \rho(Z+s) = \Pi(Z+s)\mathrm{LR}(Z+s) = r+s$, so $\rho(Z) = r$.

The argument never asks whether X, Y or Z break even: (P) locates the price point even when the cash line equals 1 everywhere else.

(b) Let $X, Y \in D$ be comonotone with $q = \Pi(X)+\Pi(Y) \neq 0$. Then $\Pi(X+Y) = q \neq 0$, so $X+Y \in D$, and $\mathrm{LR}(X+Y) = \tilde{\rho}(X+Y)/q = (\tilde{\rho}(X)+\tilde{\rho}(Y))/q = (\Pi(X)\mathrm{LR}(X)+\Pi(Y)\mathrm{LR}(Y))/q$. Both $\tilde{\rho}$ and the implied risk measure are cash additive (the former by additivity with constants and $\tilde{\rho}(c) = \Pi(c)\mathrm{LR}(c) = c$) and agree on D, hence everywhere. □

Lemma 7.6 (Law invariance of the implied pair)**.** *Assume* (P)*,* (N)*,* (CM) *and* (LI)*. Then ρ and Π are law invariant.*

Proof. If $X \overset{d}{=} Y$ then $X+c \overset{d}{=} Y+c$ for every c, so by (LI) the cash lines of X and Y have the same finite values and the same non-finite locations. Hence they have the same price point, $\Pi(X) = \Pi(Y)$, and by Proposition 4.13(b) $\rho(X) = \rho(Y)$. □

Lemma 7.7 (Homogeneity and Lipschitz continuity)**.** *Assume* (P), (N), (CM), *and let* $F \in \{\rho, \Pi\}$. *If* (RC) *or* (RM) *holds, F is positively homogeneous and Lipschitz; under* (RM) *it is* 1*-Lipschitz.*

Proof. By Lemma 7.2, ρ and Π are uniformly continuous under (RC), and monotone under (RM). Since X is comonotone with itself, $F(nX) = nF(X)$ by Lemma 7.5, so $F(qX) = qF(X)$ for rational $q > 0$. *Under* (RM)*:* $X \le Y + \|X - Y\|$ gives $F(X) \le F(Y) + \|X - Y\|$ by monotonicity and cash additivity, so F is 1-Lipschitz, and homogeneity extends to real $\lambda > 0$. *Under* (RC)*:* continuity gives $F(\lambda X) = \lambda F(X)$ for real $\lambda > 0$. Choose $\delta > 0$ with $|F(X') - F(Y')| \le 1$ whenever $\|X' - Y'\| \le \delta$, and apply it to tX, tY with $t = \delta / \|X - Y\|$; this gives $|F(X) - F(Y)| \le \|X - Y\|/\delta$. □

Lemma 7.8 (Layer formula)**.** *Assume* (P), (N), (CM), (LI) *and* (RC) *(or* (RM)*). Define, for $P(A) = p$, $h(p) := \rho(\mathbf{1}_A)$ and $g(p) := \Pi(\mathbf{1}_A)$. These are well defined by Lemma 7.6, defined for every $p \in [0,1]$ because the space is atomless, and $h(0) = g(0) = 0$, $h(1) = g(1) = 1$. For every chain $C_1 \supset \cdots \supset C_m$, $x_0 \in \mathbb{R}$ and $a_i \ge 0$,*

$$\rho\Big(x_0 + \sum_i a_i \mathbf{1}_{C_i}\Big) = x_0 + \sum_i a_i\, h(P(C_i)), \tag{7}$$

and likewise for Π with g.

Proof. The partial sum $S_{j-1} = \sum_{i<j} a_i \mathbf{1}_{C_i}$ attains its maximum on C_j, so it is comonotone with $\mathbf{1}_{C_j}$. Apply Lemma 7.5 and cash additivity successively, with $\rho(a\mathbf{1}_C) = a\rho(\mathbf{1}_C) = a\,h(P(C))$ by Lemma 7.7. The values at 0 and 1 come from $\rho(0) = \Pi(0) = 0$ and $\rho(1) = \Pi(1) = 1$. □

Lemma 7.9 (Bounded variation)**.** *Under the assumptions of Lemma 7.8, $h, g \in \mathscr{H}_1$. Under* (RM) *they are non-decreasing.*

Proof. Under (RM). For $p \le q$ choose nested $A \subset B$ with $P(A) = p$, $P(B) = q$; then $\mathbf{1}_A \le \mathbf{1}_B$ gives $h(p) \le h(q)$, and likewise for g.

Under (RC). Let L be a Lipschitz constant of ρ. Take $0 = p_0 < \cdots < p_n = 1$ and a uniform random variable U. Put $I_j = \{1 - p_j < U \le 1 - p_{j-1}\}$ and $\delta_j = h(p_j) - h(p_{j-1})$. For $\sigma \in \{0,1\}^n$ let $W = \sum_j 2(n-j)\mathbf{1}_{I_j}$ and $Y_\sigma = W + \sum_j \sigma_j \mathbf{1}_{I_j}$. The values $2(n-j) + \sigma_j$ are strictly decreasing in j, so W and Y_σ have the same upper sets $\{U > 1 - p_j\}$, of probability p_j. By (7) and summation by parts, $\rho\big(\sum_j v_j \mathbf{1}_{I_j}\big) = \sum_j v_j \delta_j$ for every non-increasing $v \ge 0$. Hence $|\sum_j \sigma_j \delta_j| = |\rho(Y_\sigma) - \rho(W)| \le L\|Y_\sigma - W\| \le L$. Choosing $\sigma_j = \mathbf{1}\{\delta_j > 0\}$ and then $\sigma_j = \mathbf{1}\{\delta_j < 0\}$ gives $\sum_j |\delta_j| \le 2L$. □

Lemma 7.10 (Representation)**.** *Under the assumptions of Lemma 7.8, $\rho = I_h$ and $\Pi = I_g$.*

Proof. For a simple $X = x_0 + \sum_i a_i \mathbf{1}_{C_i}$ with $a_i \ge 0$ and $C_i = \{X \ge x_i\}$, a direct computation gives $I_h(X) = x_0 + \sum_i a_i h(P(C_i)) = \rho(X)$. Simple positions are uniformly dense in L^∞, and ρ and I_h are both Lipschitz (Lemma 7.7 and $|I_h(X) - I_h(Y)| \le \mathrm{TV}(h)\|X - Y\|$), so $\rho = I_h$. Likewise $\Pi = I_g$. □

Proof of Theorems 5.1 and 5.2. (i)⇒(ii). By Lemma 7.1, (P), (N) and (CM) hold. Lemmas 7.3–7.10 then give $\Pi = I_g$ and $\rho = I_h$ with $h, g \in \mathscr{H}_1$, non-decreasing under (RM). On $D = \{\Pi \ne 0\} = \{I_g \ne 0\}$, $\mathrm{LR} = \rho/\Pi = I_h/I_g$, and off D the loss ratio is not finite. Under (S), at a book with $I_g(X) = 0$ the label is $\star$ exactly when the residue $\rho(X) = I_h(X)$ vanishes, which is the value of $[I_h(X) : 0]$.

(ii)⇒(i). Let LR be as in (ii). Since $I_g(X + c) = I_g(X) + c$, the cash line of X is non-finite exactly at $c = -I_g(X)$, so (P) holds and $\Pi = I_g$. (N) holds because $I_h(c) = I_g(c) = c$. Next, I_k is comonotonic additive for $k \in \mathscr{H}$: for comonotone X, Y, any two upper sets $\{X \ge a\}$, $\{Y \ge b\}$ are almost surely nested (otherwise comonotonicity fails on a $P \otimes P$-positive set); for simple comonotone X, Y the union of their upper sets is therefore a chain, giving $X = x_0 + \sum a_i \mathbf{1}_{C_i}$, $Y = y_0 + \sum b_i \mathbf{1}_{C_i}$ with $a_i, b_i \ge 0$, and additivity follows from the layer computation; for general X, Y approximate by the comonotone simple pairs $\lfloor nX \rfloor / n$, $\lfloor nY \rfloor / n$ and use the Lipschitz bound. Lemma 7.5(b) with $\tilde{\rho} = I_h$ gives (CM) and $\rho = I_h$. Hence ($\mathsf{P}_\exists$) and (R) hold by Lemma 7.1. (LI) holds because I_k depends on X only through $x \mapsto P(X > x)$.

By Lemma 7.2 the two sure amounts in (RC) and (RM) are $I_g(X)$ and $I_h(X)$, so (RC) holds because I_h, I_g are Lipschitz, and (RM) when h, g are non-decreasing. □

Proof of Propositions 5.3 and 5.4. $g(p) = \Pi(\mathbf{1}_A)$ is minus the price point of $\mathbf{1}_A$, and $h(p) = \rho(\mathbf{1}_A)$ is given by Proposition 4.13(b); both are determined by LR, including when $h = g$. The identities for D and N follow from $\Pi = I_g$, $\rho = I_h$. Monotonicity of ρ is equivalent to monotonicity of h because I_h is monotone if and only if h is non-decreasing. Proposition 5.4 collects Lemmas 7.3–7.5 and 7.7, the Lipschitz bound for I_k, and $\mathrm{LR}(\lambda X) = \lambda\rho(X)/(\lambda\Pi(X)) = \mathrm{LR}(X)$ on D (and $\lambda D = D$). □

8 Nonnegative losses

Insurance losses are nonnegative and premiums are positive. On $\mathscr{X} = L^\infty_+ \setminus \{0\}$ only upward shifts $X + c$, $c \geq 0$, are available, and the price point $-\Pi(X) < 0$ lies outside the domain, so that (P) cannot be used; nor can ($\mathsf{P}_\exists$) and (R), which pool books after moving them to zero premium. The axioms of this section are therefore stated with reals attached to the cash lines; restating them in terms of values of LR_0 alone, as in Section 4, is left open. In this section the loss ratio is a map $\mathrm{LR}_0 : \mathscr{X} \to \mathbb{R}$, and $N_0 := \{X \in \mathscr{X} : \mathrm{LR}_0(X) = 1\}$.

8.1 Axioms

Definition 8.1 (Axioms (A0)–(A5))**. (A0)** N_0 has empty interior in $(\mathscr{X}, \|\cdot\|_\infty)$.

(A1) For each $X \in \mathscr{X} \setminus N_0$ there are reals a_X, b_X with $\mathrm{LR}_0(X + c) = (a_X + c)/(b_X + c)$ for all $c \geq 0$. By Lemma 8.2, $a_X \neq b_X$, the pair is unique and $b_X > 0$; put $\rho(X) := a_X$ and $\Pi(X) := b_X$ on $\mathscr{X} \setminus N_0$.

(A2) LR_0 is law invariant.

(A3) ρ and Π are non-decreasing on $\mathscr{X} \setminus N_0$. Variant (A3$_s$): ρ and Π are uniformly continuous on $\mathscr{X} \setminus N_0$.

(A4) Let $\bar\rho, \bar\Pi$ be the continuous extensions of ρ, Π to $\mathscr{X}$ (they exist under (A3) or (A3$_s$) by Lemma B.2). For comonotone $X, Y \in \mathscr{X}$:

(A4a) $\bar\Pi(X + Y) = \bar\Pi(X) + \bar\Pi(Y)$;

(A4b) $\bar\Pi(X + Y)\mathrm{LR}_0(X + Y) = \bar\Pi(X)\mathrm{LR}_0(X) + \bar\Pi(Y)\mathrm{LR}_0(Y)$.

(A5) LR_0 is norm-continuous on $\mathscr{X}$.

Axiom (A0) excludes $\mathrm{LR}_0 \equiv 1$, for which ρ and Π are not identified, and ensures that ρ, Π are defined on a dense set. Axiom (A1) is the counterpart of Proposition 4.13(a) on a half line: since the price point is not observed, the premium must be extrapolated from $c \geq 0$, which requires the functional form to be assumed rather than derived. Axiom (A4b) is the pooling identity (CM), and (A4a) states that premiums add under comonotone pooling; on $\mathscr{X}$ the two must be stated separately. Axiom (A5) excludes jumps of LR_0 where the premium distortion vanishes; it excludes, for example, value-at-risk-type premium distortions $g = \mathbf{1}_{(a,1]}$, and it implies $g > 0$ (Step C5 in Appendix B).

Lemma 8.2. *If $X \notin N_0$ satisfies (A1) with (a, b), then $a \neq b$, the pair is unique, $b > 0$, and for every $c_0 \geq 0$ one has $X + c_0 \notin N_0$ with $\rho(X + c_0) = \rho(X) + c_0$ and $\Pi(X + c_0) = \Pi(X) + c_0$.*

8.2 Results

Theorem 8.3. LR_0 *satisfies (A0)–(A5) if and only if* $\mathrm{LR}_0 = C_h/C_g$ *on $\mathscr{X}$ for distortion functions $h \neq g$ with $g > 0$ on $(0, 1]$. In that case (h, g) is unique, also among pairs in $\mathscr{H}_1^2$ with $C_{g'} > 0$ on $\mathscr{X}$; moreover $\bar\rho = C_h$, $\bar\Pi = C_g$ and $N_0 = \mathscr{X} \cap \{C_h = C_g\}$.*

Theorem 8.4. *Axioms (A0)–(A2), (A3$_s$), (A4) and (A5) hold if and only if* $\mathrm{LR}_0 = C_h/C_g$ *with $h, g \in \mathscr{H}_1$, $h \neq g$, and $g > 0$ on $(0, 1]$.*

Proposition 8.5. $\mathrm{LR}_0 = C_h/C_g$ *with distortion functions* h, g *and* $g > 0$ *on* $(0,1]$ *if and only if* LR_0 *is the restriction to* $\mathscr{X}$ *of an extended loss ratio satisfying the axioms of Theorem 5.2 with* $\Pi > 0$ *on* $\mathscr{X}$*. The representing pair* (h,g) *of such an extension is unique if and only if* $\mathrm{LR}_0 \not\equiv 1$*. In that case the extension is unique up to the labels* $\infty, \star$ *at its non-finite values, and unique under convention* (S).

Theorem 8.3 is the loss-ratio analogue of the characterization of Wang et al. (1997). Unlike that result, the pair (ρ, Π) is defined only off N_0, and positivity of g is derived from (A5). The proofs are given in Appendix B, and Section 9 shows that none of (A2)–(A5) can be dropped.

9 Independence of the axioms

For each of the axioms ($\mathsf{P}_\exists$), (R), (LI), (RC) and (RM) we give a loss ratio that satisfies the remaining axioms and is not of the form in Theorem 5.1 or 5.2. Two facts shorten the verifications. First, by Proposition 4.5, a ratio satisfies ($\mathsf{P}_\exists$) and (R) exactly when it has the properties (P), (N) and (CM). Second, for a ratio $[A:B]$ of cash-additive functionals the sure amounts that make X unpriced and loss-free are $B(X)$ and $A(X)$ (Section 4.7), so (RC) and (RM) reduce to continuity and monotonicity of A and B. Table 4 summarizes the examples.

Dropped	**Witness**	**Why it is a witness**
($\mathsf{P}_\exists$)	$\mathrm{LR} \equiv 1$	No value is non-finite; (R), (LI), (RC), (RM) hold vacuously.
(R)	$[\max(I_{\sqrt{t}}, I_{\min(1,1.5t)}) : \mathbb{E}]$; also $\psi \circ [I_h : I_g]$, $\psi(t) = t^3$ (Example 9.3), and $[\Pi' : \Pi']$	Implied risk measure (resp. premium) is cash additive but not comonotonic additive; the cube has poles of order three.
(LI)	$[\mathbb{E}_Q : \mathbb{E}]$, $Q \neq P$ with bounded density	All other axioms hold; $\mathbb{E}_Q$ is not law invariant.
(RC) (Thm. 5.1)	$[I_h : \mathbb{E}]$ with h continuous, not of bounded variation	Uniform continuity fails; h is not of bounded variation.
(RM) (Thm. 5.2)	$[I_h : \mathbb{E}]$, $h(t) = t + \frac{1}{4}\sin 2\pi t$	h is not monotone, so no distortion pair represents it.

Table 4: Witnesses for the axioms of Theorems 5.1 and 5.2. Here $\Pi' = \max(I_{\sqrt{t}}, I_{\min(1,1.5t)})$.

Example 9.1 (Independence of ($\mathsf{P}_\exists$))**.** The constant ratio $\mathrm{LR} \equiv 1$ takes no non-finite value, so ($\mathsf{P}_\exists$) fails. Axioms (R), (RC) and (RM) hold vacuously, since their hypotheses require a non-finite value or a zero, and (LI) is clear. It is not of the form in Theorem 5.1: every such ratio is non-finite at $X = 0$, since $I_g(0) = 0$.

Example 9.2 (A loading on sure losses violates (R))**.** Fix $\kappa > 0$, $\kappa \neq 1$, and let $\mathrm{LR} = [I_h : \kappa I_g]$ with distortions h, g. The cash line of X is $(I_h(X) + c)/(\kappa(I_g(X) + c))$, non-finite only at $c = -I_g(X)$, so ($\mathsf{P}_\exists$) and (P) hold with $\Pi = I_g$, and with these weights the pooling identity (CM) holds because I_h and I_g are comonotonic additive. But $\mathrm{LR}(c) = 1/\kappa \neq 1$, so (N) fails and, by Proposition 4.5, so does (R). Directly: for unpriced X, $s(\mathrm{LR}(X+s) - 1) = (I_h(X) + s(1-\kappa))/\kappa$ depends on s. The normalization on sure losses is therefore built into (R): it is what excludes loadings on sure losses. (We conjecture, but have not proved, that an analogue of Theorem 5.1 with a known loading κ holds with $\mathrm{LR} = [I_h : \kappa I_g]$.)

Example 9.3 (Ordinal information is not enough)**.** Let $h \neq g$ be distortions and $\mathrm{LR}' = \psi \circ [I_h : I_g]$ with $\psi(t) = t^3$ and ψ fixing the non-finite values. Since ψ maps $\mathbb{R}$ to $\mathbb{R}$, LR' has the same price points as $[I_h : I_g]$, so ($\mathsf{P}_\exists$) and (P) hold with $\Pi = I_g$; (N) holds because $1^3 = 1$; (LI) holds, and so does the mediant inequality, because ψ is increasing. But LR' is not of the form in Theorem 5.1: at a book X with $I_h(X) \neq I_g(X)$ (for instance an indicator at a point where $h \neq g$), the cash line $\big((I_h(X)+c)/(I_g(X)+c)\big)^3$ has a pole of order three, while every cash line in Theorem 5.1 has a simple pole. Hence (CM) fails,

and so does (R) by Proposition 4.5. Since ψ maps 0 to 0, 1 to 1 and only 1 to 1, and keeps the non-finite values, the sure amounts that make a book unpriced or loss-free are the same for LR$'$ as for $[I_h : I_g]$, namely $I_g(X)$ and $I_h(X)$; so (RC) and (RM) hold as well, and the example satisfies every axiom except (R). Betweenness-type, ordinal information determines the loss ratio only up to an increasing transformation, as for Chew (1983) and Dekel (1986); the cardinal axiom (R) cannot be avoided.

Example 9.4 (Independence of (R))**.** Let LR $= [\rho : \mathbb{E}]$ with $\rho = \max(I_{\sqrt{t}}, I_{\min(1,1.5t)})$. ($\mathsf{P}_\exists$) and (P) hold with $\Pi = \mathbb{E}$, (N) holds, and the implied risk measure is ρ, a maximum of cash-additive maps; (LI), (RM) and (RC) hold. Take $E \subset B$ with $P(B) = 0.5$, $P(E) = 0.1$ and $X = \mathbf{1}_B + \mathbf{1}_E$. Then $\rho(\mathbf{1}_B) = \max(0.707, 0.75) = 0.75$ and $\rho(\mathbf{1}_E) = \max(0.316, 0.15) = 0.316$, with sum 1.066, while $\rho(X) = \max(0.707 + 0.316,\ 0.75 + 0.15) = 1.023$. So ρ is not comonotonic additive and (CM) fails by Lemma 7.5(a); hence (R) fails.

The ratio $[\Pi' : \Pi']$ with $\Pi' = \rho$ shows more: it equals 1 wherever it is finite, and satisfies ($\mathsf{P}_\exists$), (LI), (RC) and (RM) (both sure amounts equal $\Pi'(X)$, which is monotone and 1-Lipschitz). It fails (R) only through the clause that a pool of unpriced books is unpriced: for the books above, $\mathbf{1}_B - \Pi'(\mathbf{1}_B)$ and $\mathbf{1}_E - \Pi'(\mathbf{1}_E)$ are unpriced but their sum is not: its premium is $1.023 - 1.066 = -0.043$. Hence no axiom that depends on LR only through its finite values can replace (R) (Proposition 4.3).

Example 9.5 (Independence of (LI))**.** Let $Z = dQ/dP$ be bounded, non-constant, $Z \geq 0$ and $\mathbb{E}Z = 1$, and LR $= [\mathbb{E}_Q : \mathbb{E}]$. (P) holds with $\Pi = \mathbb{E}$, (N) holds, (CM) holds by Lemma 7.5(b) because $\mathbb{E}_Q$ and $\mathbb{E}$ are additive, so ($\mathsf{P}_\exists$) and (R) hold, and (RM), (RC) hold. But $\mathbb{E}_Q(\mathbf{1}_A) = Q(A)$ is not a function of $P(A)$, so the implied risk measure is not of the form I_h. Without (LI) the class is exactly that of Theorem 5.7.

Example 9.6 (Independence of (RC) in Theorem 5.1)**.** Let $h(t) = t\big(1 + \frac{1}{2}\sin(1/t)\big)\big/\big(1 + \frac{1}{2}\sin 1\big)$, continuous with $h(0) = 0$, $h(1) = 1$, and not of bounded variation (Figure 5) ($\int |h'|$ diverges at 0). Let LR $= [I_h : \mathbb{E}]$. (P), (N) and (LI) are clear; I_h is norm continuous (survival functions converge off a countable set, and dominated convergence applies) and additive on simple comonotone positions by the layer computation, hence comonotonic additive, so (CM) holds, and with it ($\mathsf{P}_\exists$) and (R). The implied risk measure is I_h, and by Lemma 7.9 (RC) would force h to be of bounded variation. So (RC) fails and no pair in $\mathscr{H}_1^2$ represents LR.

Remark 9.7 (Some regularity is indispensable). (P), (N), (CM) and (LI) alone do not force continuity. Using a Hamel basis one obtains $\rho = \mathbb{E} + \varphi \circ I_k$ with $k(t) = t(1-t)$ (so that I_k vanishes on constants and is cash invariant) and φ additive but discontinuous. Then $[\rho : \mathbb{E}]$ satisfies (P), (N), (CM) and (LI), hence ($\mathsf{P}_\exists$) and (R), and is not of the form in Theorem 5.1. This witness uses the axiom of choice; all the others are constructive.

Example 9.8 (Independence of (RM) in Theorem 5.2)**.** Let $h(t) = t + \frac{1}{4}\sin 2\pi t \in \mathscr{H}_1$ (Figure 5), so $h(0.4) \approx 0.547 > h(0.6) \approx 0.453$, and LR $= [I_h : \mathbb{E}]$. All other axioms hold ((RC) by the Lipschitz bound), but the implied risk measure I_h is not monotone, and no distortion pair represents LR because the pair is unique.

9.1 Witnesses for the actuarial axioms

Axioms (A0) and (A1) define ρ and Π on $\mathscr{X} \setminus N_0$, through which the remaining axioms are stated, so they are not treated as independent axioms. Two examples illustrate the identification they provide. If $\mathrm{LR}_0 \equiv 1$, (A0) fails and $(\mathrm{id}, \mathrm{id})$ and $(\sqrt{\cdot}, \sqrt{\cdot})$ both represent LR_0. If $\mathrm{LR}_0 = (C_h/C_g)^3$ with distortions $h \neq g$ and $g > 0$, then N_0 is that of C_h/C_g, but (A1) fails, because the cash line has a pole of order three at $-C_g(X)$ for $X \notin N_0$.

Remark 9.9 (Atomlessness). Atomlessness is used to attain every $p \in [0,1]$ as $P(A)$. On a space with atoms, (h, g) is identified only on the range of P.

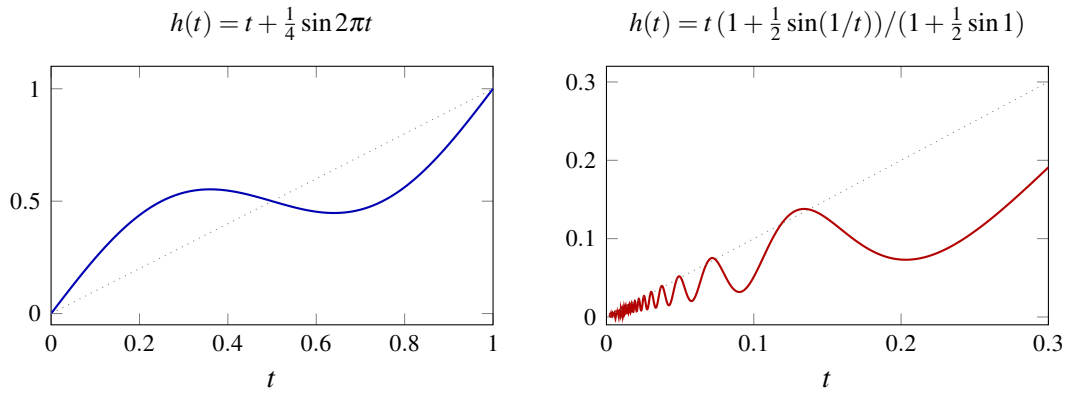


Figure 5: Numerator distortions of the witnesses for (RM) (left: of bounded variation, not monotone) and for (RC) (right, near 0: continuous, not of bounded variation). The dotted line is the identity.

Dropped	**Witness on $\mathscr{X}$**	**Reason**
(A2)	$\mathbb{E}_Q/\mathbb{E}$	N_0 lies in a closed hyperplane, so (A0) holds; not law invariant.
(A3)	$C_h/\mathbb{E}$, $h = t + \frac{1}{4}\sin 2\pi t$	$\rho(\mathbf{1}_A) > \rho(\mathbf{1}_B)$ for $\mathbf{1}_A \le \mathbf{1}_B$, $P(A) = 0.4, P(B) = 0.6$.
(A3$_s$)	$C_h/\mathbb{E}$, h continuous non-BV	Extensions exist; h is not of bounded variation.
(A4a)	$\mathbb{E}/\max(C_{\sqrt{t}}, C_{\min(1,1.5t)})$	Premium is not comonotonic additive (numbers above); (A4b) holds.
(A4b)	$\max(C_{\sqrt{t}}, C_{\min(1,1.5t)})/\mathbb{E}$	$\bar{\Pi} = \mathbb{E}$ additive, but $\bar{\Pi}\mathrm{LR}_0$ is not.
(A5)	C_h/C_g with $g = (2t-1)^+$, $h = g^2$, and $\mathrm{LR}_0 := 1$ where $C_g = 0$	LR_0 jumps: $\mathrm{LR}_0(\mathbf{1}_B + n^{-1}\mathbf{1}_C) = \frac{1}{2}$ while $\mathrm{LR}_0(\mathbf{1}_B) = 1$ for $P(B) \le \frac{1}{2}$, $C \supset B$, $P(C) = \frac{3}{4}$.

Table 5: Witnesses for (A2)–(A5) on the actuarial domain. In each case all other axioms hold.

10 Consequences

Throughout this section the axioms of Theorem 5.1 and the labelling convention (S) hold, $\mathrm{LR} = [I_h : I_g]$, and for $\beta \neq 1$ we put

$$\rho_\beta := \frac{\rho - \beta\Pi}{1-\beta}, \qquad k_\beta := \frac{h - \beta g}{1-\beta} = g + \frac{h-g}{1-\beta} \in \mathscr{H}_1.$$

Corollary 10.1 (Association). *(a) $\rho_\beta = I_{k_\beta}$ is cash additive of slope 1, and $\rho_\beta(X) = 0 \iff \mathrm{LR}(X) \in \{\beta, \star\}$. Hence $\mathrm{LR}(X) = \beta \iff \rho_\beta(X) = 0$ holds exactly off $\{\rho = \Pi = 0\} = \{\mathrm{LR} = \star\}$; it fails, for instance, at $X = 0$ and at $X - \rho(X)$ for every $X \in N$.*

(b) For $X \notin N$: $\mathrm{LR}(X - m) = \beta \iff m = \rho_\beta(X)$, and then $\Pi(X) - m = (\Pi(X) - \rho(X))/(1-\beta) \neq 0$.

(c) For $X \in N$: $\mathrm{LR}(X - m) \in \{1, \star\}$ for every m.

(d) ρ_β is $\mathrm{TV}(k_\beta)$-Lipschitz; in Theorem 5.2, $\mathrm{TV}(k_\beta) \le (1 + |\beta|)/|1 - \beta|$.

(e) ρ_β is monotone iff k_β is non-decreasing. If $h \neq g$, ρ_β is not *monotone for all β in a punctured neighbourhood of 1.*

Proof. (a) $\rho_\beta(X) = 0 \iff \rho(X) = \beta\Pi(X)$; if $\Pi(X) \neq 0$ this says $\mathrm{LR}(X) = \beta$, and if $\Pi(X) = 0$ it says $\rho(X) = 0$, i.e. $\mathrm{LR}(X) = \star$. Conversely $\mathrm{LR}(X) = \star$ means $\rho = \Pi = 0$. At $X \in N$, $Y = X - \rho(X)$ has $\rho(Y) = \Pi(Y) = 0$, so $\mathrm{LR}(Y) = \star \neq \beta$ while $\rho_\beta(Y) = 0$. (b) $\mathrm{LR}(X - m) = \beta \iff \rho - m = \beta(\Pi - m) \iff m = \rho_\beta(X)$, and $\Pi - \rho_\beta = (\Pi - \rho)/(1-\beta) \neq 0$ off N. (c) On N the cash line equals 1 away from its price point (Proposition 4.13(d)). (d) Lemma 7.7 and the Lipschitz bound for I_k. (e) I_k is monotone iff k is non-decreasing. Since $h - g$ vanishes at 0 and 1 and is non-constant, there are $p < q$ with $(h-g)(p) > (h-g)(q)$ and $p' < q'$ with $(h-g)(p') < (h-g)(q')$. As $\beta \to 1$, $(1-\beta)k_\beta =$

$h - \beta g \to h - g$ uniformly, so for β near 1 both strict inequalities persist for k_β, whatever the sign of $1-\beta$. □

Part (e) is the counterpart of Proposition 4.23 for the associated family: even when ρ and Π are distortion premia, ρ_β is in general a signed Choquet integral.

Example 10.2. Let $h = \mathrm{id}$ and $g = \sqrt{\cdot}$. Then $k_\beta(p) = (p - \beta\sqrt{p})/(1-\beta)$ and $k'_\beta(p) = \big(1 - \beta/(2\sqrt{p})\big)/(1-\beta)$. For $\beta < 1$, k_β is non-decreasing if and only if $\beta \le 2\sqrt{p}$ for all $p \in (0,1]$, that is, $\beta \le 0$; for $\beta > 1$ it is non-decreasing if and only if $\beta \ge 2\sqrt{p}$ for all p, that is, $\beta \ge 2$. Hence ρ_β is a distortion premium if and only if $\beta \notin (0,2)$; for $\beta \in (0,1) \cup (1,2)$ it is a signed Choquet integral that is not monotone (Figure 6).

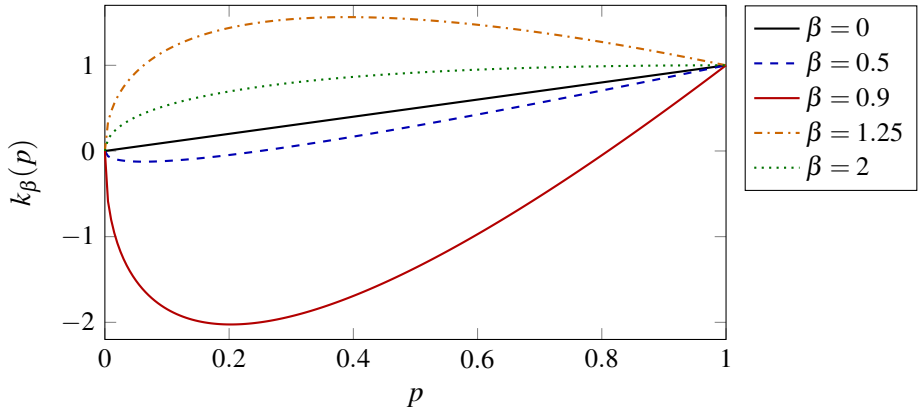


Figure 6: The distortions k_β of the associated family in Example 10.2 ($h = \mathrm{id}$, $g = \sqrt{\cdot}$). They are monotone for $\beta = 0$ and $\beta = 2$ and not monotone for $\beta \in (0,1) \cup (1,2)$.

Corollary 10.3 (Pooling at non-finite values)**.** *Assume the axioms of Theorem 5.1 and* (S)*. Then for all comonotone X, Y,*

$$\mathrm{LR}(X+Y) = [\rho(X) + \rho(Y) : \Pi(X) + \Pi(Y)].$$

In terms of loss ratios and implied premiums alone:

Case	***Pooled loss ratio***
$X, Y \in D$, $\Pi(X) + \Pi(Y) \neq 0$	*relative-cost-weighted combination* ((CM))
$X, Y \in D$, *premiums cancel*	∞ *if* $\mathrm{LR}(X) \neq \mathrm{LR}(Y)$; $\star$ *if* $\mathrm{LR}(X) = \mathrm{LR}(Y)$
$\Pi(X) = 0$, $Y \in D$	$\mathrm{LR}(Y) + \rho(X)/\Pi(Y)$, *with* $\rho(X) = \lim_{c \to 0} c\,\mathrm{LR}(X + c)$
$\Pi(X) = \Pi(Y) = 0$	∞ *unless* $\rho(X) = -\rho(Y)$, *in which case* $\star$

Proof. ρ and Π are comonotonic additive (Lemma 7.5) and $\mathrm{LR} = [\rho : \Pi]$ under (S) (Theorem 5.1). The table spells out $[\rho(X) + \rho(Y) : \Pi(X) + \Pi(Y)]$, using $\rho = \Pi\,\mathrm{LR}$ on D and Proposition 4.13(c) for books with zero premium. If premiums cancel, $\rho(X) + \rho(Y) = \Pi(X)(\mathrm{LR}(X) - \mathrm{LR}(Y))$. □

The third row has a direct reading: a loss that is carried without premium adds its size, relative to the premium base, to the loss ratio of the book it joins.

Corollary 10.4 (Orderings, convexity and shape within the class)**.** *(a)* Endpoints. *ρ is monotone iff h is non-decreasing; ρ is convex iff h is concave (Wang et al., 2020a, Thm. 3); ρ is coherent iff h is concave and non-decreasing, in which case ρ is a mixture of expected shortfalls ES_α over levels $\alpha \in [0,1]$, with the convention $\mathrm{ES}_1 := \operatorname{ess\,sup}$ (Kusuoka, 2001; Jouini et al., 2006; Föllmer and Schied, 2016) and Wang et al. (2020b, Thm. 5); the endpoint is needed, for instance, for $h = \mathbf{1}_{(0,1]}$, which gives $\rho = \operatorname{ess\,sup}$. The same holds for Π and g.*

(b) Ordering. $\mathrm{LR} \le 1$ *on* $\{\Pi > 0\}$ *iff* $h \le g$.

(c) Convex acceptance cones. *For* $\beta \neq 1$ *let* $A_\beta := \{\rho \le \beta\Pi\} = \{I_{h-\beta g} \le 0\}$. *Then* A_β *is convex iff* $h - \beta g$ *is concave, and all* A_β, $\beta \in [0,\infty) \setminus \{1\}$, *are convex iff* h *is concave and* g *is convex.*

(d) Quasi-convexity and quasi-concavity (sufficient conditions). *If* h *is concave and* g *is convex, then* $\{\Pi > 0\}$ *is convex,* $\mathrm{LR} \ge 1$ *there, and* LR *is quasi-convex on it. If, in Theorem 5.2 with* $g > 0$, h *is convex and* g *is concave, then* $\mathrm{LR} \le 1$ *on* $\mathscr{X}$ *and* LR *is quasi-concave on* $\mathscr{X}$. *In particular the actuarial ratio* $\mathbb{E}/C_g$ *with* g *concave is quasi-*concave.

(e) Rigidity. ρ_β *is convex for all* $\beta \in [0,\infty) \setminus \{1\}$ *iff* $h = g$ *and* g *is concave.*

Proof. (a) The monotonicity criterion is in Wang et al. (2020a), see also Wang et al. (2020b, Prop. 2), and the convexity criterion is Wang et al. (2020a, Thm. 3). (b) If $h \le g$ then $g - h \ge 0$ with $(g-h)(1) = 0$, so $I_{g-h} \ge 0$ and $\mathrm{LR} \le 1$ where $\Pi > 0$. If $h(p) > g(p)$ for some p, then $\mathrm{LR}(\mathbf{1}_A + c) = (h(p)+c)/(g(p)+c) > 1$ for large c, where the premium $g(p)+c$ is positive. (c) $I_{h-\beta g}$ has cash slope $1-\beta \neq 0$, so all its sublevel sets are translates of A_β; hence convexity of A_β is quasi-convexity of $I_{h-\beta g}$, which is equivalent to concavity of $h - \beta g$ (Wang et al., 2020a, Thm. 3). Taking $\beta = 0$ and $\beta \to \infty$ in $(h - \beta g)/\beta$ gives the second statement. (d) If h is concave and g is convex then I_g is concave, so $\{I_g > 0\}$ is convex; $h \ge \mathrm{id} \ge g$, so $\mathrm{LR} \ge 1$; and for $\beta \ge 1$ the set $\{I_h - \beta I_g \le 0\}$ is convex because $h - \beta g$ is concave. If h is convex and g concave, then $h \le \mathrm{id} \le g$, and $I_h - tI_g$ is concave for $t \ge 0$, so $\{\mathrm{LR} \ge t\}$ is convex; for $t < 0$ the superlevel set is all of $\mathscr{X}$, since $I_h \ge 0$ and $I_g > 0$ on $\mathscr{X}$. (e) Convexity of $\rho_\beta = I_{k_\beta}$ for $\beta < 1$ means $h - \beta g$ concave, and for $\beta > 1$ means $h - \beta g$ convex. Letting $\beta \to 1^{\mp}$ gives $h - g$ concave and convex, hence affine with zero endpoints, i.e. $h = g$; then $k_\beta = g$ and ρ_β is convex iff g is concave. The converse is clear. □

In particular, the actuarial ratio $\mathbb{E}/C_g$ with g concave is quasi-concave and not, in general, quasi-convex.

Example 10.5. Let A, B be disjoint with $P(A) = 0.2$, $P(B) = 0.3$, and $X_t = (1-t)\mathbf{1}_A + t\mathbf{1}_B$, $t \in [0,1]$. For $\mathrm{ES}_{0.9}/\mathbb{E}$ (Example 5.6), $\mathrm{ES}_{0.9}(X_t) = \max\{1-t, t\}$ and $\mathrm{LR}(X_t) = \max\{1-t,t\}/(0.2+0.1t)$, which equals 5 at $t = 0$, $10/3$ at $t = 1$ and 2 at $t = \frac{1}{2}$: the ratio lies below the larger endpoint value, as quasi-convexity requires. For $\mathbb{E}/C_{\sqrt{\cdot}}$ the ratio equals $\sqrt{0.2} \approx 0.447$ at $t = 0$, $\sqrt{0.3} \approx 0.548$ at $t = 1$ and $\sqrt{0.5} \approx 0.707$ at $t = \frac{1}{2}$: it lies above the smaller endpoint value, as quasi-concavity requires (Figure 7).

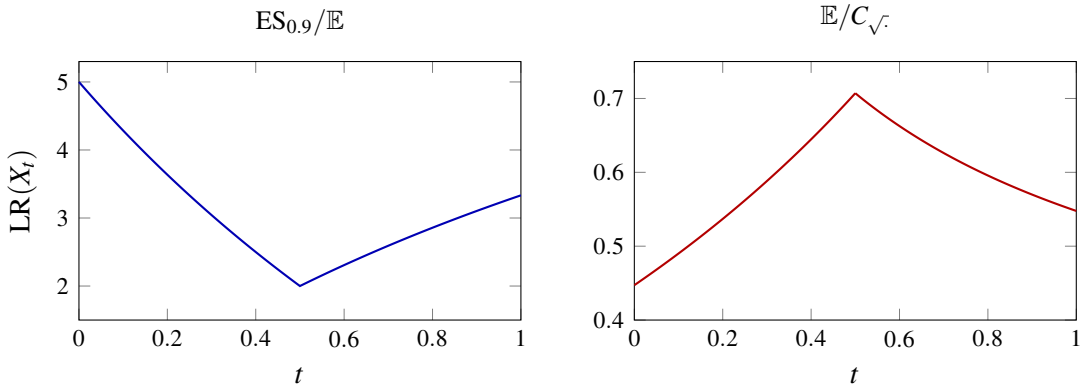


Figure 7: Loss ratios along the segment $X_t = (1-t)\mathbf{1}_A + t\mathbf{1}_B$ of Example 10.5: quasi-convex for $\mathrm{ES}_{0.9}/\mathbb{E}$ (left) and quasi-concave for $\mathbb{E}/C_{\sqrt{\cdot}}$ (right).

Example 10.6 (Further special cases)**.** (1) *Wang–Young–Panjer pairs* on nonnegative losses: Theorem 8.3. (2) *Coherent pairs*: h, g concave non-decreasing; ρ_β is a signed mixture of expected shortfalls. (3) *Actuarial ratio* $\mathbb{E}/C_g$, $h = \mathrm{id}$: Proposition 5.3 gives $g(p) = p/\mathrm{LR}(\mathbf{1}_A)$ for $P(A) = p$, so the premium

distortion is identified from the loss ratios of Bernoulli books. (4) *VaR-type pairs* are admissible in Theorem 5.2; ρ_β is non-monotone near $\beta = 1$; a VaR-type premium $g = \mathbf{1}_{(a,1]}$ is excluded from Theorem 8.3, since $g > 0$ fails. (5) *Gini-type ratios*: $\mathrm{LR} - 1 = I_{h-g}/I_g$ is a location-invariant signed Choquet "variability" divided by a premium (compare Bellini et al., 2022).

11 Discussion and open problems

Summary. A loss ratio carries its own premium and its own risk measure. The premium is the price point of the cash line, the sure payment at which the book changes hands, and the risk measure is loss ratio times premium, visible even where the ratio explodes as the residue of the singularity. With this reading, a single substantive axiom, comonotone pooling at relative cost, together with law invariance and regularity of the price point and the zero of the cash lines, characterizes the loss ratios that are ratios of two Choquet-type functionals. The characterization is sharp, and the pair is unique. At the level of representations it is a transfer of the theory of Yaari (1987), Schmeidler (1986), Wang et al. (1997) and Wang et al. (2020a); at the level of the loss ratio it is new, and it comes with facts that have no single-functional counterpart: no non-trivial loss ratio is monotone (Proposition 4.23), the association holds exactly off $\{\rho = \Pi = 0\}$ and is signed (Corollary 10.1), and pooling extends consistently to books with infinite or indeterminate loss ratios (Corollary 10.3).

Open problems.

(1) *Beyond comonotone pooling.* The most promising direction is to drop (R) in favour of an axiom that allows diversification, so that the implied premium and risk measure are convex or subadditive rather than comonotonic additive. The implied premium and the implied risk measure are defined without (R) only partly (the premium needs only (P); the accounting reading of ρ needs pooling with sure losses), and the resulting class, a ratio of Kusuoka-type functionals, has no characterization yet. Replacing comonotonic additivity by convexity leads to suprema of signed Choquet integrals (Liu et al., 2020), so no single pair (h, g) and no uniqueness should be expected.

(2) *Weaker cardinal axioms.* Can (R) be replaced by the mediant inequality plus a supplementary condition? By Section 9, such a condition must involve non-finite values of LR. The inequality implies additivity for comonotone pairs strictly on the same side of break-even, but not for pairs straddling 1 or lying in the break-even set.

(3) *Order properties of* LR. (RM) is an order property of LR, but of the sure amounts at which its cash lines become non-finite or vanish. Whether it is equivalent to an order property of the values of LR alone is open; by Proposition 4.23 it cannot be pointwise monotonicity.

(4) *Loadings on sure losses.* We conjecture that an analogue of Theorem 5.1 holds for ratios $[I_h : \kappa I_g]$ with a known loading $\kappa > 0$, which would admit constant loss ratios $\neq 1$.

(5) *Quasi-convexity.* Necessity of the conditions in Corollary 10.4(d) is open.

(6) *Actuarial domain.* Whether (A4) imposed only on pairs with $X, Y, X + Y \notin N_0$ suffices is open; a truncation-continuity (Greco-type) version of the signed tier on the cone is not known.

(7) *Unbounded losses.* Extensions to L^p and Orlicz spaces via Wang et al. (2020b) are not treated.

12 The use of AI: process, verification and a reporting standard

This section is the second half of the paper. It reports how the result was obtained with AI, at a level of detail that lets a reader judge how far the result can be trusted, and it is meant to be read as a worked example of the reporting standard proposed in Section 12.9. The tools are identified in the declaration before the references. The section describes the organization of the work, how the models acquire

knowledge and the risks this entails for mathematical work, the errors found, the guardrails used, the formal check in Lean, and what has and has not been checked.

12.1 Organization of the work

The work used Claude (Anthropic) through the Claude desktop application in Cowork mode, which runs the assistant in a sandboxed environment and can coordinate several AI agents in a workflow. AI agents worked on designated parts of the project, under the author's direction. A main assistant organized the work and drafted and revised the manuscript. A roundtable of six agents, five of them each assigned a strand of the literature and one acting as chair, surveyed the axiomatic theory of premiums, risk measures and performance ratios and produced a synthesis. A larger panel of agents covered ten literature strands, proposed candidate statements, ranked them by means of judge agents, drafted proofs, and subjected the proofs to separate AI-assisted checking runs over several rounds. The literature review was conducted by these AI assistants under the author's supervision: the author defined the strands to be covered and the questions to be answered, and selected and framed the material presented in Section 2. Some agents were given the names of researchers as personas, to represent the perspective of a literature strand; these were simulations, and the researchers concerned did not participate, were not consulted and bear no responsibility for the content. The research questions, the choice of formulation and all decisions on the content of the paper were made by the author.

Figures 8 and 9 show the organization of the work. In Figure 8 the author assigns the task of each AI component and reviews its output. Figure 9 shows the internal pipeline of the panel, in which candidate statements and proofs pass through ranking and repeated critical review before being returned to the author. Figure 10 gives a schematic picture of the panel: the ten strand agents around a shared digest, the proof bench of judge, prover and checking agents, and the author's place in the loop.

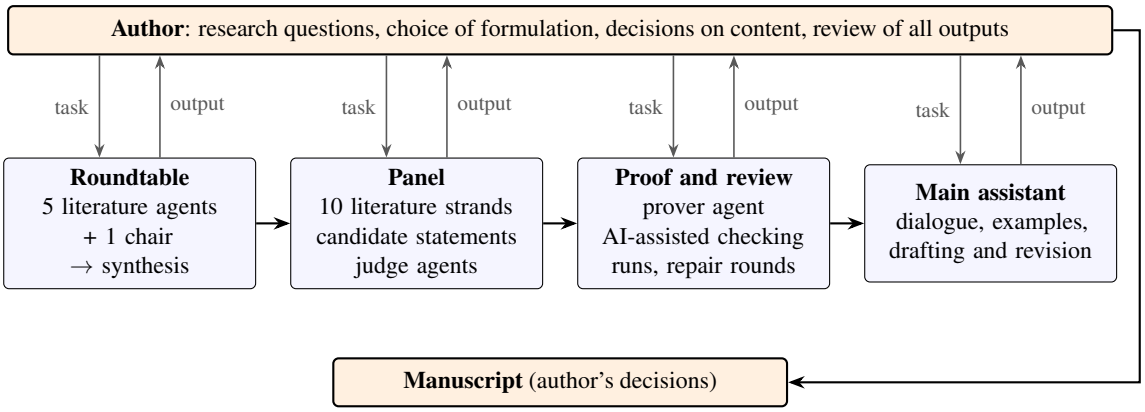


Figure 8: Organization of the work. The author assigns tasks to each AI component and reviews every output; the manuscript reflects the author's decisions. The stages were used in the order shown, with returns to earlier stages when the author redirected the work.

AI assisted with the following tasks:

- literature exploration;
- formulation of mathematical statements and drafting of proofs;
- construction of counterexamples and examples;
- manuscript drafting and revision.

12.2 How the knowledge base for this paper was assembled

The agents draw on two kinds of knowledge: what their underlying models learned in training, which can be recalled only approximately, and documents read into their working context during the task. In

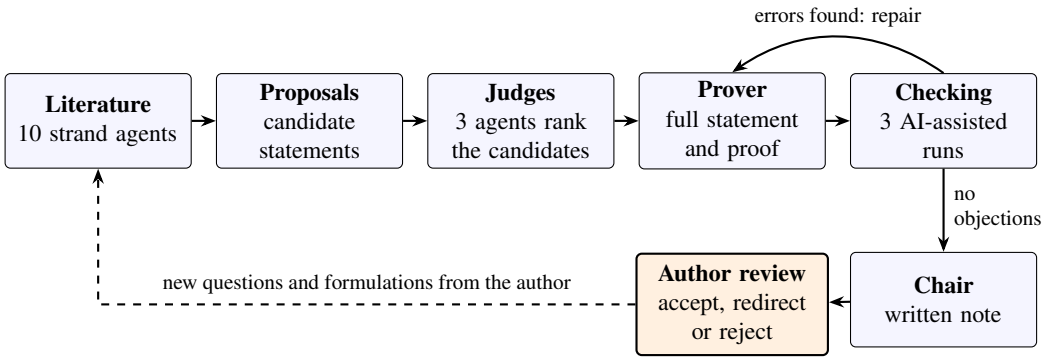


Figure 9: Pipeline of the panel. Literature agents survey their strands; proposal agents formulate candidate statements, which judge agents rank. A prover agent drafts a full statement and proof, which separate AI-assisted checking runs examine; reported errors are sent back for repair over several rounds. A chair agent writes up the result for the author, who accepts it, redirects the work or rejects it. Agreement among checking runs does not establish correctness.

this project the second kind was organized as follows.

The five strands of the roundtable were identified by the author: premium principles and capital allocation; conic finance and performance ratios; coherent risk measures and acceptance sets; axiomatic risk measures and elicitability; and the extended-loss-ratio framework with its association. Its synthesis served as the starting point of the panel.

The panel divided the literature into ten strands: insurance-pricing axiomatics; Choquet integrals and comonotonic additivity; signed Choquet integrals; law-invariant coherent and convex risk measures; acceptability indices and quasi-convex duality; return and star-shaped risk measures; elicitability; performance-ratio axiomatics; decision theory and functional equations; and a critique of the earlier synthesis. Each strand agent returned exact statements of if-and-only-if theorems with their hypotheses, a confidence label (high, medium or low) and a note on its source. For example, the signed representation results of Cerreia-Vioglio et al. (2012, 2015) were taken from their working paper versions.

There was no separate retrieval index. Instead, the strand reports were merged into a written digest of about 415,000 characters, which, together with the ten candidate formulations (about 245,000 characters), formed the shared knowledge base of the later stages: the judge and prover agents were instructed to read these files in full before working, and Section 2 was written from the digest. The confidence labels therefore travelled with each statement into the stages that used it.

In addition, the author used Gemini (Google) to prepare summaries of the key works cited in Section 2: Gerber (1974), Yaari (1987) and Wang et al. (1997) on premium principles; Murofushi et al. (1994), Cerreia-Vioglio et al. (2012, 2015) and Wang et al. (2020a) on signed Choquet integrals; Artzner et al. (1999) and Kusuoka (2001) on coherent risk measures; and Cherny and Madan (2009), Chew (1983) and Dekel (1986) on performance ratios and ratio representations. Each summary stated the core results of the work and their role in this paper. The descriptions in Section 2 and the uses of these works in Sections 4, 5, 9 and 10 were compared with these summaries; no discrepancy was found, and three descriptions in Section 2 were made more precise. Since the summaries are themselves AI output, this comparison is a consistency check, not a check against the original publications.

Later, the author supplied author versions of three cited papers: Wang et al. (2020a) (working paper of June 2019), Wang et al. (2020b) (version of May 2020, with a correction of May 2023) and Han et al. (2025) (version of June 2024). The statements taken from them were checked against these versions: the definition of I_h, the characterization theorem used in the representation step and the convexity theorem of Wang et al. (2020a); the monotonicity criterion and Lipschitz bound, which are also stated there; the extension to general spaces, the convention $\mathrm{ES}_1 = \operatorname{ess\,sup}$ and the mixture representation of Wang et al. (2020b); and the axioms and the two characterization theorems of Han et al. (2025). No discrepancy

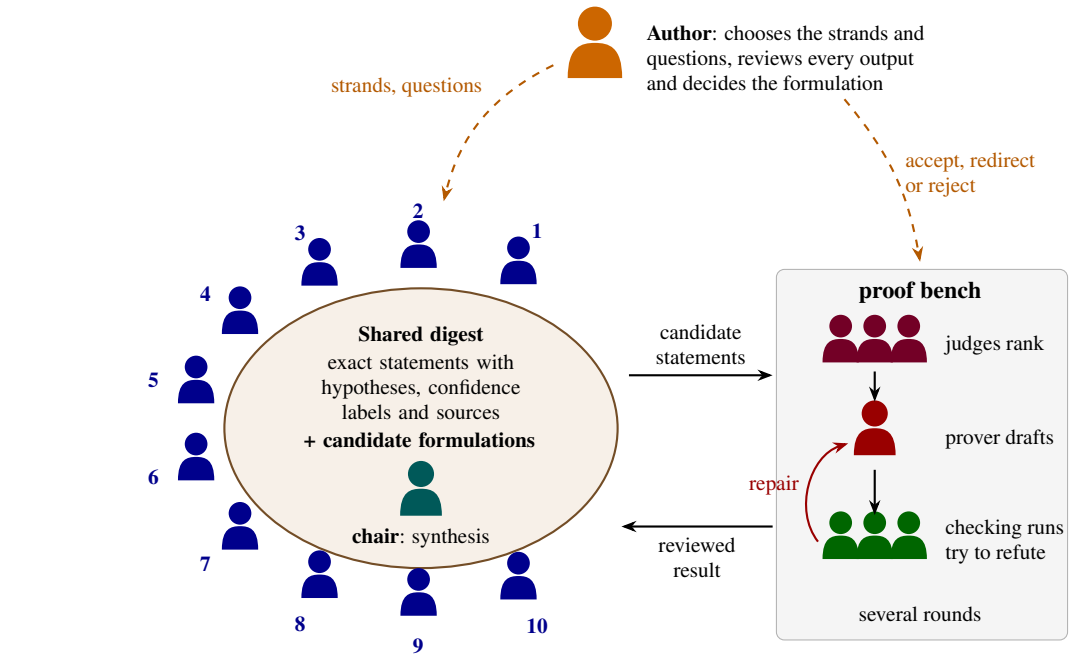


Figure 10: The panel at work. Ten literature agents (numbered as in the key), each assigned a strand by the author, write exact statements into a shared digest, which together with the candidate formulations forms the knowledge base read in full by the later agents; a chair agent synthesizes. Candidate statements pass to the proof bench, where judge agents rank them, a prover agent drafts proofs and separate checking runs try to refute them, with errors returned for repair. The author sets the strands and questions and accepts, redirects or rejects each result. The icons are schematic: the agents are separate runs of a language model.

was found. Lemma numbers that could differ between these versions and the published articles were removed, and the description of Han et al. (2025) in Section 2 was rewritten from the paper.

12.3 Hallucinations and semantic errors encountered

Statements about the literature. The main risk of hallucination concerned the exact hypotheses, conclusions and numbering of cited results. Two entries of the digest disagreed on whether the mixing measure in the Kusuoka representation includes an atom at the endpoint; the paper therefore states the mixture over $[0, 1]$ with an explicit convention (Corollary 10.4(a)) and cites the representation without theorem numbers. For the same reason, the representation step used in the proofs is not taken from the literature but reproved (Lemmas 7.7–7.10).

Errors in mathematical content. Table 6 lists the errors in AI-generated mathematics that were found during the project. Most are semantic: the statement proved, or claimed, differed from the statement intended, typically at a degenerate case (zero premium, the break-even set, the labels ∞ and $\star$) or in the scope of a claim.

Error	Found by	Resolution
$\mathbb{E}/C_g$ with concave g claimed quasi-convex (early synthesis)	panel agents	quasi-concave; Cor. 10.4(d)
quasi-convexity stated as an if-and-only-if in four candidate formulations	judge agent ($h=t$, $g=\sqrt{t}$, $\beta=2$)	only sufficient conditions kept
candidate route via Sincov and Cauchy equations	judge agent (continuous h not of bounded variation)	route abandoned
main argument claimed to apply verbatim when (ρ,Π) is only assumed to exist	checking run	separate argument given
association claimed to hold exactly on the finite domain	checking run	holds off $\{\rho=\Pi=0\}$; Cor. 10.1(a)
"$h-g$ concave" as a sufficient condition for quasi-convexity	main assistant, during rewriting	h concave, g convex
converse of Theorem 5.1 failed for (LI) (labels unconstrained)	checking run	(LI) compares finite values only
claimed equivalence with the cash-line formulation	checking run	convention (S) added; Remark 4.19
regularity axioms undefined for two counterexamples	checking run	(RC), (RM) stated for the price point and the zero of the cash lines
uniqueness of the extension in Prop. 8.5	review with ChatGPT	unique up to labels, unique under (S)
independence claimed for (A0) and (A1)	review with ChatGPT	treated as illustrations; Sec. 9
mixture of expected shortfalls missing the endpoint	review with ChatGPT	$\mathrm{ES}_1 := \operatorname{ess\,sup}$; Cor. 10.4(a)
Revision of Section 12.7		
implied risk measure identified before (CM) was derived (circular citation)	adversarial agent of the panel	identified after Step 3; Lemma 7.1
regularity axioms in a multiplied form excluded the witness of Example 9.3	two adversarial agents	(RC), (RM) use only the predicates "not finite", "$=0$", "$=1$"
test of "$\mathrm{LR}(W+c)=1$ for all $c\neq 0$" by sampling c gave false results on break-even books	adversarial agent	finite predicate $\mathrm{LR}(W+1)=1$
failure of (R) for $[\Pi':\Pi']$ claimed from random trials, which never hit the pole	adversarial agent	explicit books; Example 9.4
non-finite point of $[I_h:I_g]$ misplaced in the example for ($\mathsf{P}_\exists$)	main assistant, on rereading	non-finite at $X=0$

Table 6: Errors in AI-generated mathematics found during the project, and their resolution.

12.4 Guardrails applied in this project

- *Labels on knowledge.* Every literature statement in the digest carried a confidence label and a note on its source, as described above.
- *Independent summaries.* Summaries of the key cited works prepared with Gemini were compared with their descriptions and uses in the paper.
- *Self-contained proofs.* Results whose exact statements were uncertain are reproved; in particular the representation of comonotonic additive functionals, for which the proof cites Wang et al. (2020a, Thm. 1), is reproved in Lemmas 7.7–7.10.
- *Adversarial checking.* The three checking runs on the panel's proofs were instructed to refute; one tested the degenerate case $\mathrm{LR}\equiv 1$ with a premium that is not comonotonic additive, and two recomputed the numbers in the counterexamples. The restructured formulation of Sections 4–7 was given to a further checking run with instructions to look for circularity and for the cases $\Pi(X)=0$, break-even books, $\mathrm{LR}\equiv 1$ and $c=0$.

- *Repair loop.* Reported gaps were returned to the prover agent; the panel's proofs went through two repair rounds before the final checking round.
- *Counterexamples.* Each axiom is tested against an explicit loss ratio (Section 9).
- *Recomputation.* The values in Examples 5.5, 5.6, 10.5 and in the counterexamples were recomputed by script, and the monotonicity range in Example 10.2 was confirmed on a grid.
- *Formal checking.* The loss-ratio-specific steps were formalized in Lean and compiled; see Section 12.6.
- *A second system.* The review with ChatGPT, a model from a different provider, found three of the errors in Table 6 that the earlier checks had missed.

12.5 What the prover produced and what was formally checked

The "prover" of Figure 9 is a language-model agent that writes proofs in ordinary mathematical language; it is not a formal theorem prover. In this project it drafted the first complete if-and-only-if theorem, in a formulation based on an assumed cash-line axiom, with proofs of the results that are now Theorems 5.1, 5.2, 5.7, 8.3 and 8.4 and of the appendix lemmas; the final checking round of these proofs returned two reports without objections and one report of gaps in side claims, which were repaired. The present formulation through the implied premium and risk measure (Lemmas 7.3–7.6 and Proposition 4.13) was drafted afterwards by the main assistant and checked by one further run, whose findings appear in Table 6.

The loss-ratio-specific steps were also checked formally in Lean (Section 12.6).

12.6 Formal checking in Lean

Lean and what a formal check establishes

Lean 4 (de Moura and Ullrich, 2021) is an interactive proof assistant: a programming language in which definitions, statements and proofs are written in a formal language based on dependent type theory, and in which every proof is checked by a small trusted program, the kernel. A proof may be written step by step or with the help of tactics, procedures that search for parts of a proof automatically, but whatever a tactic produces is checked again by the kernel. Mathlib (The mathlib Community, 2020) is the community library of formalized mathematics for Lean; it supplies the real numbers, groups, limits and the other notions used below.

Two features of Lean are used to state precisely what has been checked. A proof can be left unfinished by writing `sorry`, which Lean accepts with a warning; a finished development contains none. The command `#print axioms` lists the foundational axioms on which a theorem depends; ordinary classical mathematics in Lean uses exactly three of them, `propext` (propositional extensionality), `Classical.choice` (the axiom of choice) and `Quot.sound` (soundness of quotients), and an unfinished proof shows up in this list as `sorryAx`.

A successful check establishes that the formal statements follow from the formal definitions. It does not establish that the formal definitions and statements say what the paper says, which must be checked by a human reader, and it says nothing about results that enter as hypotheses or are not formalized.

For this paper a formal check is useful for three reasons. First, the proofs were drafted with AI assistance, and the errors recorded in Table 6 show that such drafts can contain steps that read well and are wrong; the kernel does not depend on how a proof was produced. Second, the steps that are specific to loss ratios (Lemmas 7.3–7.6) are elementary but delicate, with case distinctions at $\Pi(X) = 0$, at break-even books and at the price point, which is where informal checking is most likely to miss a case. Third, formalization forces the hypotheses to be stated exactly, and here it shows that those steps use only a few algebraic properties of comonotonicity and equality in law (see "What was formalized" below).

What was formalized

The formalization works in an abstract setting. Positions form an additive commutative group $\mathscr{X}$ with an additive embedding $c \mapsto \mathrm{const}(c)$ of the sure losses, a relation "comonotone" under which every position is comonotone with every constant and which is preserved when both positions are shifted by constants, and a relation "equal in law" preserved by a common cash shift. A loss ratio is a map from $\mathscr{X}$ to $\mathbb{R} \cup \{\mathtt{none}\}$, where `none` stands for a non-finite value, so that the labels ∞ and $\star$ are identified as in (LI). The axioms read as follows; `prem` is the implied premium and `rhoD` is $\Pi \cdot \mathrm{LR}$.

```
def AxP : Prop := ∀ x : X, ∃! c : ℝ, LR (x + S.const c) = none
def AxN : Prop := ∀ c : ℝ, c ≠ 0 → LR (S.const c) = some 1
def AxLI : Prop := ∀ x y : X, S.eqd x y → LR x = LR y
def AxCM (hP : AxP S LR) : Prop :=
  ∀ x y : X, S.comon x y → LR x ≠ none → LR y ≠ none →
    prem hP x + prem hP y ≠ 0 →
    LR (x + y) = some ((rhoD hP x + rhoD hP y) / (prem hP x + prem hP y))
```

These coincide with Axioms 4.6, 4.9, 4.10 and 4.20, with one difference of form: `AxCM` is stated with the products $\Pi \cdot \mathrm{LR}$ over the sum of the premiums, which is Axiom 4.10 written as in its second expression, so that the relative-cost weights $w_{X,Y}$ do not appear in the Lean text. The file has about 500 lines with 10 definitions and 26 lemmas and theorems. The axioms ($\mathsf{P}_\exists$) and (R) of Section 4, their equivalence with (P), (N) and (CM) (Lemma 7.1), and the regularity axioms in the form of Lemma 7.2 are not yet formalized. Table 7 lists the results of the paper that are formalized.

Result in the paper	**Lean name**
Implied premium, uniqueness and cash additivity (Lemma 7.3)	`prem`, `prem_shift`, `prem_const`
Implied risk measure and cash additivity (Lemma 7.4)	`rho`, `rho_cash`
Möbius form and residue (Proposition 4.13(a)–(c))	`mobius`, `rho_eq`, `residue`
Comonotonic additivity, also on the break-even set (Lemma 7.5)	`comon_additive`
Law invariance of the implied pair (Lemma 7.6)	`LI_transfer`
Converse construction and uniqueness of the pair	`converse`, `unique_pair`
Association (Corollary 10.1(a))	`association`
Pooling at non-finite values (Corollary 10.3)	`pooling_extended`
Algebraic core of Proposition 4.23	`no_monotone_core`
Theorem 5.1, given the representation step	`main_conditional`

Table 7: Results formalized in Lean.

The theorem `main_conditional` takes the representation step as a hypothesis: any cash-additive, comonotonic additive, law-invariant and regular functional is of the required form (Wang et al., 2020a, Thm. 1). From (P), (N), (CM) and (LI) it derives that the implied premium and the implied risk measure satisfy the hypotheses of that step and that $\mathrm{LR} = [\rho : \Pi]$ everywhere. Not formalized are the representation step itself (Lemmas 7.7–7.10), which would require a theory of Choquet integrals that Mathlib does not provide, the fact that L^∞ with comonotonicity and equality in law is an instance of the abstract setting, the layer formula, and the examples and counterexamples.

How the check was carried out

The Lean file was written by the main assistant. Lean could not be installed in the assistant's cloud workspace, whose network policy blocks the servers that distribute Lean, so the file was compiled on the author's computer. The author installed Lean and created the project; the assistant read the build output from the project folder through the link between the Claude desktop application and the author's

computer, wrote corrected versions of the file into the same folder, and the author ran the next build (Figure 11).

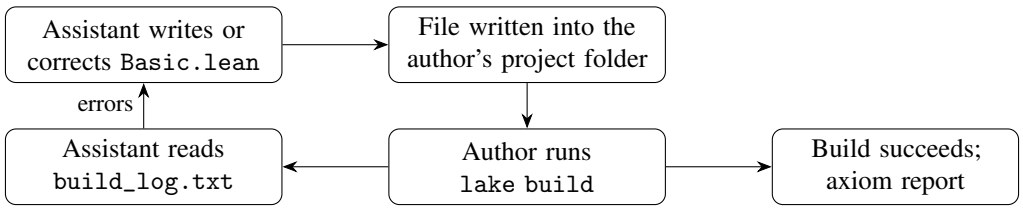


Figure 11: The build-and-repair loop used for the Lean check.

Table 8 gives the timeline. Times are in UTC and are taken from the session records and from the modification times of the files in the project folder.

Time (UTC)	Step
21 Oct, 21:38	The assistant writes a first Lean file for the results in Table 7; it cannot be compiled in the cloud workspace.
24 Oct, 17:08	The author asks how a formal proof check can be used; the assistant describes what is feasible (the loss-ratio steps), what is hard (the representation step) and how to install Lean.
24 Oct, 17:26–17:31	The author installs the Lean extension for the code editor and the Lean toolchain manager.
24 Oct, 17:35–17:37	The author creates the project `loss_ratio_lean` with Mathlib; Lean and Mathlib version 4.36.0-rc1; the precompiled Mathlib files are downloaded.
24 Oct, 17:38–17:39	The assistant inspects the project configuration, confirms that it is complete, and places the Lean file in it, replacing the template placeholder.
25 Oct, 12:47–12:53	First attempt to build: the command `lake` is not found, because the toolchain had not been added to the terminal's search path; fixed by loading the toolchain environment.
25 Oct, 12:55	Build 1: four errors, none in a definition or statement. One proof stopped one algebraic step short; in three proofs Lean's simplifier rewrote $\mathrm{const}(-c)$ as $-\mathrm{const}(c)$, so that cash additivity no longer applied.
25 Oct, 12:57	The assistant writes a corrected file and appends `#print axioms` for ten main theorems.
25 Oct, 13:04	Build 2: one error, in the converse construction, where the goal was displayed in an unexpanded form. The axiom report shows the three standard axioms for nine theorems and `sorryAx` for the converse.
25 Oct, 13:05	The assistant adds one line that states the goal explicitly.
25 Oct, 13:08	Build 3 succeeds. All ten theorems depend only on the three standard axioms.
25 Oct, 13:09	The assistant compares the Lean definitions of (P), (N), (CM) and (LI) and the theorem statements with Section 4.
25 Oct, 13:37	At the author's request, the project's own build output is deleted and the file is rebuilt from scratch: success in 8.8 s, with the same axiom report.

Table 8: Timeline of the Lean check.

Checks performed

The following checks were made to ensure that the outcome is what is reported.

- *No change to statements during repair.* The corrections after builds 1 and 2 changed proof steps only; no definition and no theorem statement was altered.
- *No unfinished proofs.* The final file contains no `sorry`, no `admit` and no added `axiom`.

- *Axiom report.* `#print axioms` for each of the ten theorems in Table 7 lists only `propext`, `Classical.choice` and `Quot.sound`.
- *Clean rebuild.* The final build started from an empty build directory, so its result does not rely on cached output.
- *Identity of the file.* The file compiled in the final build was compared with the file reported here and found identical.
- *Fidelity of statements.* The formal axioms and theorem statements were compared with the paper by the assistant; this comparison is to be repeated by the author before the code is released.

The remaining warnings concern style and the use of renamed library lemmas; they do not affect correctness. The Lean source and the build log are available in the repository `https://github.com/hirbod-assa/loss-ratio-lean`, which contains the Lean project (Lean and Mathlib version 4.36.0-rc1) together with instructions for rebuilding it with `lake build`.

12.7 A revision: axioms on the loss ratio alone

In an earlier version, the pooling axiom (CM) and the regularity axioms were stated through the implied premium and the implied risk measure. A reviewer remarked that some axioms were therefore based on derived quantities, and that axioms stated on the loss ratio itself would be preferable. This subsection records how the present formulation, in which every axiom refers only to values of LR (Section 4.1), was obtained.

Exploration by the author. With the main assistant, the author first tried to replace the implicit premium by explicit formulas in terms of LR: the premium as the limit of $c(1-\mathrm{LR}(X+c))$ for $|c|\to\infty$ divided by $1-\mathrm{LR}(X)$, a two-quote formula on the cash line, and, for books that break even, the limit of the premium of $X+\varepsilon Y$ as $\varepsilon\to 0$. The main assistant checked each proposal. All of them agree with the implied premium where they are defined, but on books that break even, and for $h=g$, the finite values of the cash line are all equal to 1 and the formulas are $0/0$ (Proposition 4.15); the perturbation repairs this only if the premium is already known to be continuous. An intermediate version stated (CM) with weights given by the points at which the cash lines are not finite. The author rejected it: such a point is the premium under another name, and the requirement was that no axiom use the premium, a risk measure or a point read off a cash line, but only values of LR.

The panel. The author then asked for a panel to replace (CM). It was run as a workflow of nine agents. Four proposal agents worked from different angles: stability of level sets under comonotone pooling together with a condition on the shape of the cash lines; a single pooling identity on cash lines; elimination of the weight from (CM); and the limits of what a pure loss-ratio axiom can achieve. Each proposal was passed to its own adversarial agent, instructed to refute it and to recompute its numbers. All four survived with repairs; the errors the adversarial agents found are in the last part of Table 6. A judge agent ranked the four, chose the proposal that became ($\mathsf{P}_\exists$) and (R), adopted the form of the regularity axioms from another proposal, and collected the impossibility results that became Proposition 4.3 and the witnesses of Section 9. The judge also checked the combined system in exact rational arithmetic on a four-point space for three representing pairs, a pair of distortions, $h=g$ and a signed pair: (R) held on every test pair (600 for each of the first and third, 450 for $h=g$), including, for the first, 75 pairs of non-constant break-even books, 75 mixed pairs and 150 pairs whose premiums cancel; the sure amounts in (RC) and (RM) equalled I_g and I_h; and each witness of Section 9 failed. The judge's report and its script are in `ai_process/` in the repository. The main assistant then wrote Section 4.1, Lemmas 7.1 and 7.2 and the changes in the other sections, and repeated the numerical checks.

The author's decisions. The author adopted the judge's recommendation and asked whether the hypothesis that the pooled books are unpriced could be replaced by the existence of $\lim_{s\to 0} s(\mathrm{LR}(X+s)-1)$. The answer is no (Remark 4.4), and the reason, that no condition on finite values suffices, was stated as Proposition 4.3. The nonnegative case of Section 8 was left in its earlier form.

What was checked. The proofs of Lemmas 7.1 and 7.2 and of Proposition 4.3 were checked by one adversarial agent and the judge and by computation; they are not formalized in Lean and were not part of the review with ChatGPT.

12.8 Verification status

The proofs have undergone AI-assisted critical review. Such review is fallible and does not constitute formal verification; this applies equally to the informal review with ChatGPT. No claim of machine-checked correctness is made beyond the results formalized in Lean, in the abstract setting and with the hypothesis described in Section 12.6. Independent human review of all proofs has not yet taken place; the reviewer's remark reported in Section 12.7 concerned the formulation of the axioms. Apart from the three papers named above, the references have not been checked against the original publications.

12.9 A proposed standard for reporting AI-assisted proofs

The declarations of AI use that journals now require state which tools were used and that the author takes responsibility for the result. For a paper whose proofs were drafted or checked with AI, this is not enough for a reader to judge the result. The errors in Table 6 show why: most of them are statements that are almost right, and a reader who knows only that AI was used cannot tell which parts of the paper were exposed to such errors and which checks they passed. We therefore propose that a mathematical paper in risk management or actuarial science whose results were obtained with substantial AI assistance report the items in Table 9. The table also shows where each item is reported for the present paper.

Item	**What to report**	**In this paper**
1. Tools	providers, models and versions, mode of access, dates of use	declaration
2. Division of labour	who posed the question, chose the formulation and decided on content; which tasks were given to AI	Secs. 1, 12.1
3. Workflow	the agents or sessions used, their roles and order, the review loops, and the points at which the author intervened	Figs. 8–10; Sec. 12.7
4. Knowledge base	how the literature was gathered, how reliable each statement was taken to be, and which cited statements were checked against the original works	Sec. 12.2
5. Error log	every error in AI-generated mathematics that was found, how it was found, and how it was resolved	Table 6
6. Guardrails	the checks applied: adversarial review, review by a second system, recomputation of numbers, counterexamples, self-contained proofs of uncertain citations	Sec. 12.4
7. Formal verification	which results were checked by a proof assistant, in what setting, and which steps were taken as hypotheses	Sec. 12.6
8. Verification status	for each part of the paper, whether it was reviewed by AI, by a human, or formally checked; and what has not been checked	Sec. 12.8
9. Records	the formal proofs and the intermediate records of the AI process, publicly available	code availability; `ai_process/` in the repository

Table 9: Proposed items for reporting AI-assisted proofs, and where they are reported in this paper.

Three observations from this project motivate the items. First, the errors were found by different checks: checking runs within the panel, the main assistant during rewriting, and a review with a model from a different provider, which found three errors that the earlier checks had missed (Table 6). Reporting the checks, and not only their conclusion, tells the reader which kinds of error have been looked for. Second, the decisive step, the reformulation through the implied premium and risk measure, came from the author's objection to an AI-drafted formulation that was internally correct but assumed what should

have been proved; a record of the division of labour makes such contributions visible. Third, agreement among AI reviews does not establish correctness, and a formal check covers only what was formalized; the verification status must therefore be stated part by part. The list is a first proposal. It will need to be tested on other papers, and in particular adapted to results whose proofs are checked formally in full.

A Proof of Theorem 5.7 (no law invariance)

Lemmas 7.3, 7.4, 7.5 and 7.7 do not use (LI). Put $\nu_1(A) := \rho(\mathbf{1}_A)$ and $\nu_2(A) := \Pi(\mathbf{1}_A)$. They are null-invariant, because ρ and Π act on almost-sure classes, normalized by $\rho(1) = \Pi(1) = 1$, and the layer formula (7) holds with $\nu_i(C_j)$ in place of $h(P(C_j))$, for every chain $C_1 \supset \cdots \supset C_m$.

Bounded variation. Given a chain $\emptyset = A_0 \subset \cdots \subset A_n = \Omega$, run the proof of Lemma 7.9 with $I_j := A_j \setminus A_{j-1}$; then $\{W \geq 2(n-j)\} = A_j$, and one obtains $\|\nu_1\|_{bv} \leq 2L$. Under (RM), ν_1 is monotone, i.e. a capacity.

Jordan decomposition of a game. For $\nu \in \mathscr{V}$ let $\nu^+(A) := \sup \sum_j (\nu(A_j) - \nu(A_{j-1}))^+$ over chains from $\emptyset$ to A, and $\nu^- := \nu^+ - \nu$. For $A \subset B$, appending B to a chain for A gives $\nu^+(B) \geq \nu^+(A) + (\nu(B) - \nu(A))^+$. Hence ν^+ is non-decreasing, and $\nu^-(B) - \nu^-(A) \geq (\nu(B) - \nu(A))^+ - (\nu(B) - \nu(A)) \geq 0$, so ν^- is non-decreasing. Moreover $\nu^+(A) \leq \|\nu\|_{bv}$ (append Ω), and null-invariance passes to $\nu^\pm$ (if $P(A \triangle A') = 0$, map $A_j \mapsto A_j \cap A'$ for $j < n$ and $A_n \mapsto A'$, which preserves all ν-values).

Representation. $C_{\nu^\pm}$ (with the correction term by $\nu^\pm(\Omega)$) is monotone with cash slope $\nu^\pm(\Omega)$, hence Lipschitz, and $C_\nu = C_{\nu^+} - C_{\nu^-}$ by linearity in ν. Now ρ and C_{ν_1} are Lipschitz and agree on simple positions by (7), so $\rho = C_{\nu_1}$; likewise $\Pi = C_{\nu_2}$.

Converse and uniqueness. As in the proof of Theorems 5.1 and 5.2, with ν in place of $h \circ P$; null-invariance makes C_ν well defined on almost-sure classes and makes almost-sure nestedness sufficient. Uniqueness follows because $\Pi(\mathbf{1}_A)$ is minus the price point of $\mathbf{1}_A$ and $\rho(\mathbf{1}_A)$ is given by Proposition 4.13(b).

Law invariance. If (LI) holds, Lemma 7.6 makes $\nu_i(A)$ a function of $P(A)$ alone, so $\nu_i = k_i \circ P$; since the space is atomless, $\|k \circ P\|_{bv} = \mathrm{TV}(k)$, so $k_i \in \mathscr{H}_1$. Conversely $k \circ P$ gives a law-invariant functional. □

B Proofs for the actuarial domain

Proof of Lemma 8.2. $a/b = \mathrm{LR}_0(X) \neq 1$, so $a \neq b$. *Uniqueness.* If $(a+c)(b'+c) = (a'+c)(b+c)$ for all $c \geq 0$, comparing coefficients gives $a + b' = a' + b$ and $ab' = a'b$; writing $a' = a + t$, $b' = b + t$ gives $t(a-b) = 0$, so $t = 0$. *Positivity.* If $b \leq 0$, the pole $-b \geq 0$, at which the numerator does not vanish, would contradict finiteness of LR_0. *Shifts.* $\mathrm{LR}_0(X + c_0) = (a + c_0)/(b + c_0) \neq 1$, and the profile of $X + c_0$ is the shifted profile of X. □

Lemma B.1 (Sierpiński and quantile coupling)**.** *(a) If $P(B) > 0$, $P(\cdot \mid B)$ carries a uniform random variable. (b) For every $Y \in L^\infty$ there is a uniform U with $Y = q_Y(U)$ a.s. If $X \overset{d}{=} Y$ and $X_n \to X$ in norm, then $Y_n := q_{X_n}(U)$ satisfies $Y_n \overset{d}{=} X_n$ and $\|Y_n - Y\| \leq \|X_n - X\|$.*

Proof. (a) $P(\cdot \mid B)$ is atomless; by Sierpiński's theorem there are sets $B_t \subset B$, increasing in dyadic $t \in [0,1]$, with $B_1 = B$ and $P(B_t) = tP(B)$. Put $W := \inf\{t : \omega \in B_t\}$; then $\{W < t\} \subset B_t \subset \{W \leq t\}$, so $P(W \leq t \mid B) = t$. (b) Off the atoms of Y set $U := F_Y(Y)$; on each atom $\{Y = y\}$ set $U := F_Y(y-) + (F_Y(y) - F_Y(y-))W_y$ with W_y uniform under $P(\cdot \mid Y = y)$ by (a). Then U is uniform and $q_Y(U) = Y$. If $|X_n - X| \leq \varepsilon$ a.s., quantiles move by at most ε, so $\|q_{X_n}(U) - q_X(U)\| \leq \|X_n - X\|$ and $q_X = q_Y$. □

Lemma B.2 (Extension)**.** *Assume (A0)–(A2) and (A3) (respectively (A3$_s$)). Then ρ,Π are 1-Lipschitz (respectively uniformly continuous) on $\mathscr{X}\setminus N_0$, hence extend uniquely to continuous $\bar\rho,\bar\Pi$ on $\mathscr{X}$. The extensions are cash additive for $c\geq 0$, law invariant, monotone under (A3), and $\bar\Pi\geq 0$.*

Proof. Let $X,Y\notin N_0$ and $t=\|X-Y\|$. Then $Y+t\notin N_0$ (Lemma 8.2) and $X\leq Y+t$, so (A3) gives $\rho(X)\leq\rho(Y)+t$; thus ρ,Π are 1-Lipschitz, and $\mathscr{X}\setminus N_0$ is dense by (A0). Cash additivity passes to the limit. Monotonicity: if $X\leq Y$, choose $X_n,Y_n\notin N_0$ within $1/n$ of X,Y; then $X_n\leq Y_n+2/n\notin N_0$, and let $n\to\infty$. Law invariance: N_0 is law invariant by (A2), and off N_0 so are ρ,Π by uniqueness in Lemma 8.2. If $X\overset{d}{=}Y$ and $X_n\to X$ with $X_n\notin N_0$, then $Y_n:=q_{X_n}(U)\notin N_0$ satisfies $\rho(Y_n)=\rho(X_n)$ and $Y_n\to Y$ (Lemma B.1), so $\bar\rho(X)=\bar\rho(Y)$. Finally $\Pi>0$ off N_0 gives $\bar\Pi\geq 0$. Under (A3$_s$) the same argument applies with extension by uniform continuity. □

Lemma B.3 (Sweep)**.** *Let k be bounded Borel with $k(0)=0$, such that C_k is comonotonic additive on $\mathscr{X}$ (true for $k\in\mathscr{H}$ and for continuous k). If $k(p_0)\neq 0$ for some $p_0\in(0,1]$, then $\mathscr{X}\cap\{C_k=0\}$ has empty interior.*

Proof. Write $X=q_X(U)$ and $A=\{U>1-p_0\}$. Then X and $\mathbf{1}_A$ are comonotone, so $C_k(X+t\mathbf{1}_A)=C_k(X)+t\,k(p_0)$ for $t>0$, which vanishes for at most one t. Since $\|X+t\mathbf{1}_A-X\|=t$, every ball around X meets $\{C_k\neq 0\}$. □

Proof of Theorem 8.3. Step C1 (law invariance, extension). Lemma B.2.

Step C2 (pole). $\bar\Pi$ is comonotonic additive by (A4a) and continuous, hence positively homogeneous. Then $\bar\Pi(2)=2\bar\Pi(1)$ and $\bar\Pi(2)=\bar\Pi(1)+1$ give $\bar\Pi(1)=1$.

Step C3 (zero). $\bar\rho$ and $\mathrm{LR}_0\bar\Pi$ are continuous (by (A5) for the second) and agree off N_0, where $a_X=\mathrm{LR}_0(X)b_X$; so $\bar\rho=\mathrm{LR}_0\bar\Pi$ on $\mathscr{X}$. By (A4b), $\bar\rho$ is comonotonic additive, hence homogeneous, and $\bar\rho(1)=1$ as in C2.

Step C4 (representation). For $P(A)=p>0$ put $h(p):=\bar\rho(\mathbf{1}_A)$, $g(p):=\bar\Pi(\mathbf{1}_A)$ and $h(0)=g(0)=0$. They are well defined (law invariance) and non-decreasing (monotonicity). Since $\varepsilon\mathbf{1}_A\leq\mathbf{1}_A$, homogeneity and monotonicity give $(1-\varepsilon)h(p)\geq 0$, so $h,g\geq 0$; thus h,g are distortion functions. A simple $X\in\mathscr{X}$ equals $\sum_i(x_i-x_{i-1})\mathbf{1}_{B_i}$ with $x_{-1}=0$, $B_i=\{X\geq x_i\}$, a comonotone sum of positive multiples of indicators, so $\bar\rho(X)=C_h(X)$; approximating X by the simple $\lceil nX\rceil/n\in\mathscr{X}$ gives $\bar\rho=C_h$ on $\mathscr{X}$, and likewise $\bar\Pi=C_g$.

Step C5 (positivity of g). (a) If $g(p)=0$ then $h(p)=\mathrm{LR}_0(\mathbf{1}_A)\cdot 0=0$. (b) By (A0) some $W\notin N_0$ exists and, by Lemma 8.2, $\bar\rho(W)\neq\bar\Pi(W)$, so $h(q_0)\neq g(q_0)$ for some q_0; as $g(q_0)\geq 0$ and $g(q_0)\neq 0$ by (a), $g(q_0)>0$. (c) Suppose $g(p)=0$ for some $p>0$ and take nested B,C with $P(B)=p$, $P(C)=q_0$ and $X_n=\mathbf{1}_B+n^{-1}\mathbf{1}_C$. Then $\bar\Pi(X_n)=g(q_0)/n>0$, $\bar\rho(X_n)=h(q_0)/n$, so $\mathrm{LR}_0(X_n)=h(q_0)/g(q_0)\neq 1$. But $\bar\Pi(\mathbf{1}_B)=g(p)=0$ forces $\mathbf{1}_B\in N_0$, so $\mathrm{LR}_0(\mathbf{1}_B)=1$, contradicting (A5) since $X_n\to\mathbf{1}_B$.

Step C6 (conclusion). $\mathrm{LR}_0=\bar\rho/\bar\Pi=C_h/C_g$. If $h=g$ then $N_0=\mathscr{X}$, contradicting (A0).

Converse. Let $\mathrm{LR}_0=C_h/C_g$ with $h\neq g$ and $g>0$ on $(0,1]$; then $C_g>0$ on $\mathscr{X}$, because $P(X>x)>0$ for $x\in[0,\operatorname{ess\,sup}X)$. (A1) holds with $a_X=C_h(X)$, $b_X=C_g(X)$; (A2)–(A5) are immediate; (A0) follows from Lemma B.3 with $k=h-g$, since $N_0=\mathscr{X}\cap\{C_{h-g}=0\}$.

Uniqueness among signed pairs. If also $\mathrm{LR}_0=C_{h'}/C_{g'}$ with $(h',g')\in\mathscr{H}_1^2$ and $C_{g'}>0$, then by Lemma 8.2 $C_{h'}=\rho$ and $C_{g'}=\Pi$ off N_0. The set N_0 is determined by LR_0 and has dense complement (Lemma B.3, $h\neq g$), so by continuity $C_{h'}=\bar\rho=C_h$ and $C_{g'}=C_g$; evaluating at indicators gives $h'=h$, $g'=g$. □

Proof of Theorem 8.4. Steps C1, C2 and the extension hold with uniform continuity in place of monotonicity. For $Z\in L^\infty$ and $M>\|Z\|$ put $\tilde\rho(Z):=\bar\rho(Z+M)-M$, independent of M by cash additivity. Then $\tilde\rho$ is law invariant, cash additive, uniformly continuous and comonotonic additive (the shifts Z_i+M are comonotone elements of $\mathscr{X}$, and Step C3 with (A4b) applies). Lemmas 7.7–7.10, applied to $\tilde\rho$ and

likewise to $\tilde{\Pi}$ (using (A4a)), give $\tilde{\rho} = I_h$, $\tilde{\Pi} = I_g$ with $h, g \in \mathscr{H}_1$. Step C5 holds verbatim: in (b) the sign $g \geq 0$ comes from $\bar{\Pi} \geq 0$, not from monotonicity, and in (c) nested sets are comonotone in either direction. The converse uses the Lipschitz bounds for (A3$_s$) and Lemma B.3 for (A0). □

Proof of Proposition 8.5. ($\Rightarrow$) Extend LR_0 to L^∞ by $[I_h : I_g]$; this satisfies Theorem 5.2, and $I_g > 0$ on $\mathscr{X}$ because $g > 0$. ($\Leftarrow$) Theorem 5.2 gives distortions h, g, and $I_g > 0$ on $\mathscr{X}$ forces $g > 0$ on $(0, 1]$ (otherwise $I_g(\mathbf{1}_A) = g(P(A)) = 0$ for some A). *Uniqueness.* If $\mathrm{LR}_0 \not\equiv 1$ then $h \neq g$, and two extensions have pairs that agree by the uniqueness in Theorem 8.3. Their finite values, I_h/I_g on $\{I_g \neq 0\}$, and their non-finite locations, $\{I_g = 0\}$, therefore coincide; the labels at the non-finite values are fixed by (S) and are otherwise arbitrary (for instance, the value at 0 may be ∞ or $\star$). If $\mathrm{LR}_0 \equiv 1$, $[I_{\mathrm{id}} : I_{\mathrm{id}}]$ and $[I_{\sqrt{\cdot}} : I_{\sqrt{\cdot}}]$ have different representing pairs: the price point of $\mathbf{1}_A$ is $-P(A)$ for the first and $-\sqrt{P(A)}$ for the second. □

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During this research and the preparation of this manuscript, the author used Claude (Anthropic), accessed through the Claude desktop application in Cowork mode between 20 and 25 October 2026, with the model identifiers `claude-sonnet-4-6`, `claude-fable-5-1`, `claude-sonnet-5-5` and `claude-opus-5-5` recorded in the available session logs, to assist with literature exploration, formulation of mathematical statements, construction and criticism of proofs and counterexamples, writing a Lean formalization of part of the proofs, and manuscript drafting and revision. The author also used ChatGPT (OpenAI) for an informal review of the manuscript's proofs, which identified potential errors and qualifications and suggested revisions to the mathematical statements and to this disclosure; this review did not constitute formal proof verification. The author used Gemini (Google) to prepare summaries of the content of key cited works, against which the descriptions of the literature in the manuscript were compared. Substantive mathematical assistance and checking procedures are described in Section 12. AI-generated arguments and references were treated as proposals requiring verification. The author takes full responsibility for the final mathematical statements, proofs, references and presentation.

Code availability

The Lean 4 formalization described in Section 12.6 is available at `https://github.com/hirbod-assa/loss-ratio-lean`.

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